Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Multi Observation Production Centre

Global Ocean 3D Chlorophyll-a concentration, Particulate Backscattering coefficient, Particulate Organic Carbon, Photosynthetic Available Radiation and Downwelling Irradiance Product

MULTIOBS_GLO_BIO_BGC_3D_REP_015_010

Issue: 5.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	April 2020	All	Initial version of the document	R. Sauzède, H. Claustre	S. Guinehut
2.0	February 2021		Updates linked to the improvement of the vertical resolution, the POC quality retrieval and the temporal extension of the product	R. Sauzède, H. Claustre	S. Guinehut
3.0	September 2021		Upgraded Chl product replacing the previous Uitz et al. 2006 parameterization method by a neural network-based method.	R. Sauzède, P.R. Renosh, J. Uitz and H. Claustre	S. Guinehut
4.0	November 2024		Improvement of Chl and b_{bp} /POC estimations thanks to new BGC-Argo data available and more accurate Chl BGC-Ago data used for the neural network training.	R. Sauzède, P.R. Renosh, C. Schmechtig, J. Uitz and H. Claustre	H. Etienne
5.0	November 2025		A major update includes new light variables: PAR, ED380, ED412, and ED490. These are derived at local time noon using SOCA-light models (Renosh et al., 2023). The SOCA-light model was enhanced in 2025 through retraining, utilizing more profiles and slightly revised input variables.	R. Sauzède, P.R. Renosh, C. Schmechtig, J. Uitz and H. Claustre	H. Etienne

ACRONYM TABLE

Acronym	Description
b_{bp}	Particulate backscattering coefficient
BGC	Biogeochemical
Chl	Chlorophyll-a concentration
ED	Downwelling Irradiance
ED380	Downwelling Irradiance at 380 nanometers
ED412	Downwelling Irradiance at 412 nanometers
ED490	Downwelling Irradiance at 490 nanometers
EMS	Eastern Mediterranean Sea
GLO	Global
HPLC	High Performance Liquid Chromatography
KD490	Diffuse Attenuation Coefficient at 490 nanometers
MAPD	Median Absolute Percent Difference
MDS	Marine Data Store
MLD	Mixed Layer Depth
MLP	Multilayer Perceptron
MULTIOBS	Multi Observation
MY	Multi Year
NN	Neural Network
NASPG	North Atlantic Subpolar Gyre
NASTG	North Atlantic Subtropical Gyre
PAR	Downwelling Photosynthetic Available Radiation
PCA	Principal Component Analysis
POC	Particulate Organic Carbon
PUM	Product User Manual
QUID	Quality Information Document
r^2	Coefficient of Determination
RMSD	Root Mean Square Difference
R_{rs}	Remote Sensing Reflectance
SeaBASS	SeaWiFS Bio-Optical Archive and Storage System
SLA	Sea-level Anomaly
SO	Southern Ocean
SOCA	Satellite Ocean Colour merged with Argo data
TAC	Thematic Assembly Center
WMO	World Meteorological Organization
WMS	Western Mediterranean Sea

KEY DEFINITIONS

Accuracy Metrics	The accuracy of the derived products is evaluated using a comprehensive suite of metrics computed on both the native and log-transformed data spaces to provide a complete picture of model performance. The RMSD is used to quantify the overall magnitude of the error in the original units, and the MAPD is used to assess the typical relative error; both of these metrics are computed directly on the non-log-transformed (true) data. In contrast, the slope and r^2 is used to measure the linearity and skill of the estimates relative to the observations, and this metric is specifically computed from the linear regression performed on the log-transformed data.
errors	The MULTIOBS_GLO_BIO_BGC_3D_REP_015_010 product provides the estimated error alongside each core biogeochemical variable (e.g., b_{bp}, POC, Chl, PAR, ED380, ED412, and ED490). The errors for all variables, including b_{bp} and Chl from the SOCA2024 retrieval, as well as PAR, ED380, ED412, and ED490 from the SOCA-light-2025 estimation, are determined using a consistent method: measuring the dispersion within the ensemble of neural network models used for the respective estimations.
L3	Level 3 data are derived geophysical variables that have been aggregated/projected onto a well-defined spatial grid over a well-defined time period
MLP	A MLP is a fundamental class of feedforward Artificial Neural Network consisting of an input layer, one or more hidden layers, and an output layer. It is fully connected and uses non-linear activation functions in its hidden layers to learn and approximate complex, non-linear relationships, typically trained using backpropagation.
Monthly Climatology	A monthly climatology is a long-term average of a specific atmospheric or oceanic variable (like Chl, POC or PAR) calculated for a particular calendar month across multiple years of historical data. It provides a stable, representative picture of the typical conditions for that month, effectively smoothing out inter-annual variability and short-term weather events to reveal underlying seasonal patterns. Climatologies are crucial as a baseline for detecting anomalies and long-term climate change trends.
Spikes	A spike in a Chl profile refers to an abrupt, short-lived, and abnormally high value recorded by a sensor on a BGC-Argo float or a similar platform.

RELEVANT DOCUMENT LIST

PUM	https://documentation.marine.copernicus.eu/PUM/CMEMS-MOB-PUM-015-010.pdf
SQO	https://documentation.marine.copernicus.eu/SQO/CMEMS-MOB-SQO-015-010.pdf

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the approach followed to produce and validate the Copernicus Marine Service global ocean product **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010**. The product provides a 3D global reconstruction at ¼° across 36 vertical levels (from the surface to 1000 m depth) for five key biogeochemical properties, which are described below:

- Particulate Organic Carbon (POC),
- Particulate backscattering coefficient (b_{bp}),
- Chlorophyll-a concentration (Chl),
- Photosynthetic Available Radiation (PAR), and
- Downwelling Irradiance at 380 nm (ED380), 412 nm (ED412) and 490 nm (ED490).

Note however that because this product uses satellite L3 ocean colour products upstream, areas with persistent sea ice or cloud coverage are not reconstructed.

Variables	Product Code	Area	Update
POC, b_{bp}, Chl, PAR, ED380, ED412, and ED490 fields	MULTIOBS_GLO_BIO_BGC_3D_REP_015_010	Global	Yearly

This product provides two datasets on a regular spatial grid (0.25°x0.25°) but with different temporal resolutions:

- **cmems_obs-mob_glo_bgc-chl-poc_my_0.25deg_P7D-m** containing the weekly fields.
- **cmems_obs-mob_glo_bgc-chl-poc_my_0.25deg-climatology_P1M-m** containing the multi-year monthly climatological fields.

The product is extended in time at least once a year so that it covers from January 1998 up to between 12 to 24 months before present. The multi-year monthly climatological dataset contains the fields generated from averaged inputs on the same time period as the available weekly fields.

The product benefits from the best quality ocean observation inputs available at the time of the release and is produced with a single system (no change in time in the setup, extended in time using the same system) to minimize the risk of quality loss or spurious signals appearing in time.

I.2 Summary of the results

(Sauzède et al., 2016) developed **a machine learning-based method** that effectively extends surface bio-optical properties (i.e., b_{bp}) downwards into deeper layers of the ocean. This method, known as **SOCA** (for Satellite Ocean Color merged with Argo data, hereafter referred to as SOCA2016), merges ocean colour observations with *in situ* hydrological properties to infer the vertical distribution of b_{bp}. SOCA2016 was trained using Biogeochemical-Argo (BGC-Argo) hydrological and b_{bp} profiles matched with satellite data. The method has since been upgraded to SOCA2020 for b_{bp} and SOCA2021 for Chl, which were used for the production from September 2021. The current product benefits from the new SOCA2024 method, which improves the estimation of both b_{bp} and Chl. This improvement is mainly due to the

availability of new BGC-Argo data and more accurate Chl data that have been refined in delayed mode (see details in Section II.2.2). This SOCA2024 method is used to generate the current high-level 3D gridded MULTIOBS_GLO_BIO_BGC_3D_REP_015_010 product of Chl, POC and b_{bp}.

The particulate organic carbon (POC) is obtained using the **particulate backscattering coefficient (b_{bp})**, a widely used bio-optical proxy for POC (Cetinić et al., 2012; Evers-King et al., 2017; Loisel et al., 2002; Stramski et al., 2008). The POC is derived from b_{bp} estimations using the POC *vs.* b_{bp} at 700 nm, $b_{bp}(700)$, relationship established for this study. This relationship is based on an *in situ* global open ocean POC database (Evers-King et al., 2017) linked to b_{bp} at 700 nm, the reference wavelength for BGC-Argo b_{bp} measurements (see details in II.2.4).

As we use the satellite L3 ocean colour product for the reconstruction of the b_{bp}, POC, and Chl vertical distributions, the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product contains gaps for areas with cloud or sea ice coverage. Moreover, as the methodology to retrieve the vertical distribution of POC and Chl was developed for open ocean waters, we applied a bathymetric mask of a minimum of 1,000 m depth to our product. The Black Sea, with its exceptionally high levels of coloured dissolved organic matter, exhibits bio-optical anomalies (Organelli et al., 2017). For this reason, we have chosen not to deliver the product for this region.

Building on the success of the original SOCA models, which demonstrated the feasibility of deriving bio-optical properties from BGC-Argo float data, we have developed an extended model, SOCA-light, to generate vertically resolved light profiles (PAR, ED380, ED412 and ED490) in the open ocean (Renosh et al., 2023). SOCA-light utilizes a unique, decade-long dataset of PAR, ED380, ED412, and ED490 measurements collected by BGC-Argo floats. In 2025, the SOCA-light model was retrained using an expanded set of profiles and slightly revised input variables. This updated version is hereafter referred to as SOCA-light-2025. By incorporating satellite ocean colour data alongside temperature and salinity profiles, the model enables the estimation of light fields at any open ocean location with available match-up data. Beginning with the November 2025 release, the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product will include four additional light variables - PAR, ED380, ED412, and ED490 maintaining spatial and temporal resolutions consistent with those of Chl, b_{bp}, and POC.

I.3 Estimated Accuracy Numbers

The accuracy of the product is assessed against independent datasets, which are not used to develop the method. These datasets correspond to 20% randomly chosen profiles of the BGC-Argo database used to develop the upgraded SOCA2024 method. POC accuracy is assessed based on the performance of b_{bp} sensors in retrieving POC from BGC-Argo profiling floats, as examined in the study by (Johnson et al., 2017). For Chl the accuracy is assessed against a global independent pigment database of reference obtained from High Performance Liquid Chromatography (HPLC) analyses performed at SAPIGH[1], the HPLC analytical service hosted at the Institut de la Mer de Villefranche (IMEV[2]). This dataset has been merged with the MAREDAT database (Peloquin et al., 2013).

The estimated accuracy numbers of the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product is given synthetically in Table 1 (at the global scale and for all depths). The statistics are computed on both true data and log-transformed data. Note that for the remainder of the document, the RMSD (Root Mean Square Difference) and MAPD (Median Absolute Percentage Deviation) are computed on non-log-

[1] SAPIGH webpage: https://lov.imev-mer.fr/web/facilities/sapigh/
[2] IMEV webpage: https://www.imev-mer.fr/web/

transformed data, while the r^2 (determination coefficient of the linear regression between estimates and observations) is computed from the linear regression performed on log-transformed data. The regional accuracy of the product is assessed through the comparison between retrieved and *in situ* values, displayed as geographical maps of the Median Percent Difference (MPD) for b_{bp} and Chl (Figure 16 and Figure *23*). Evaluation of the product's vertical structure accuracy is performed through scatterplots comparing retrieved and *in situ* values. Figure 10, Figure 15 and Figure 21 display these for b_{bp} and Chl, while Figure 24, Figure 25, Figure 31 and Figure 32 show them for light variables.

Table 1: Estimated accuracy statistics (RMSD: Root Mean Squared Difference, MAPD: Median Absolute Percent Difference, r^2: determination coefficient of the linear regression between estimates and observations) for the key biogeochemical properties considered in the product. Validation datasets: BGC-Argo (20% validation dataset) for b_{bp}, HPLC database for Chl, and SeaBASS for Light variables (PAR, ED380, ED412, ED490).

Variable	Metrics			Units
	RMSD	MAPD (%)	r^2	
b_{bp}	0.00025	11	0.84	m^{-1}
Log10(b_{bp})	0.093	1.46	0.90	m^{-1}
POC	≈20	≈30		mg m^{-3}
Chl	0.35	40	0.43	mg m^{-3}
Log10(Chl)	0.34	27	0.53	mg m^{-3}
PAR	85.48	54.39	0.75	µmol quanta m^{-2} s^{-1}
Log10(PAR)	0.46	26.81	0.86	µmol quanta m^{-2} s^{-1}
ED380	0.10	63.42	0.70	W m^{-2} nm^{-1}
Log10(ED380)	0.58	22.37	0.88	W m^{-2} nm^{-1}
ED412	0.16	62.29	0.72	W m^{-2} nm^{-1}
Log10(ED412)	0.59	28.15	0.86	W m^{-2} nm^{-1}
ED490	0.19	48.95	0.69	W m^{-2} nm^{-1}
Log10(ED490)	0.42	25.53	0.86	W m^{-2} nm^{-1}

II PRODUCTION SYSTEM DESCRIPTION

The production center name is the Institut de la Mer de Villefranche (IMEV)/Laboratoire d'Océanographie de Villefranche (LOV), identified by the code MULTIOBS-LOV-VILLEFRANCHE-FR.

Table 2: Table of product specifications

Product code	**MULTIOBS_GLO_BIO_BGC_3D_REP_015_010**
Production system's name	SOCA2024 & SOCA-light-2025
Production centre's name	Institut de la Mer de Villefranche (IMEV)/Laboratoire d'Océanographie de Villefranche (LOV) (MULTIOBS-LOV-VILLEFRANCHE-FR)
Issue	5.0
Short description	Global REP 3D particulate backscattering coefficient, particulate organic carbon and chlorophyll-a concentrations from merged satellite and hydrological data using the upgraded SOCA (SOCA2024) machine learning-based method (formerly presented in Sauzède et al., 2016) and global 3D Downwelling Photosynthetic Available Radiation, Downwelling Irradiance at 380 nm, Downwelling Irradiance at 412 nm and Downwelling Irradiance at 490 nm at local time noon using SOCA-light model (Renosh et al., 2023). The SOCA-light model underwent retraining in 2025, based on an expanded dataset of profiles and updated input variables (SOCA-light-2025).
Geographical coverage	Global with a 0.25°x0.25° spatial resolution
Variables	**bbp:** Particulate backscattering coefficient (m^{-1}) **bbp_error**: Error on the particulate backscattering coefficient (m^{-1}) **poc**: Particulate organic carbon (mg m^{-3}) **poc_error**: Error on the particulate organic carbon (mg m^{-3}) **chl**: Total chlorophyll-a concentration (mg m^{-3}) **chl_error**: Error on the total chlorophyll-a concentration (mg m^{-3}) **PAR:** Downwelling Photosynthetic Available Radiation (µmol quanta m^{-2} s^{-1}) **PAR_error:** Error on the Downwelling Photosynthetic Available Radiation (µmol quanta m^{-2} s^{-1}) **ED380:** Downwelling Irradiance at 380 nanometers (W m^{-2} nm^{-1}) **ED380_error:** Error on the Downwelling Irradiance at 380 nanometers (W m^{-2} nm^{-1}) **ED412:** Downwelling Irradiance at 412 nanometers (W m^{-2} nm^{-1}) **ED412_error:** Error on the Downwelling Irradiance at 412 nanometers (W m^{-2} nm^{-1}) **ED490:** Downwelling Irradiance at 490 nanometers (W m^{-2} nm^{-1}) **ED490_error:** Error on the Downwelling Irradiance at 490 nanometers (W m^{-2} nm^{-1}) **time:** Time of product (hours since 1950-01-01 00:00:00) **depth**: Depth (m) **longitude**: Longitude (degrees East) **latitude**: Latitude (degrees North)

Available time series	**cmems_obs-mob_glo_bgc-chl-poc_my_0.25deg_P7D-m**: weekly mean fields from January 1998 to ~12 to 24 months before present **cmems_obs-mob_glo_bgc-chl-poc_my_0.25deg-climatology_P1M-m:** multi-year monthly climatological fields (averaged on the 1998-~12 to 24 months before present time period)
Vertical resolution	36 vertical levels from 0 to 1000 m depth are spaced as follows: every 5 meters from 0 to 70 m, every 10 meters from 70 to 100 m, every 25 meters from 100 to 300 m, every 50 meters from 300 to 600 m, and every 100 meters from 600 to 1000 m.

II.1 Origin of input fields

The data sources used to develop the SOCA2024 and SOCA-light method are provided in Table 3. Satellite-derived reflectances (R_{rs}) and diffuse attenuation coefficient at 490 nm (KD490), Photosynthetically Available Radiation (PAR), Sea Level Anomaly (SLA), Mixed Layer Depth (MLD), and temperature/salinity profiles serve as input data for training the machine learning algorithm. The BGC-Argo b_{bp} and Chl profiles are the targeted outputs. Additionally, Table 4 includes the data sources used to reconstruct b_{bp} (and POC), Chl, PAR, ED380, ED412 and ED490 fields for producing **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010**. It is important to note that from the initial list of inputs, SOCA-light excludes SLA, while SOCA2024 excludes the KD490.

Beyond the predictors listed in Table 3 and Table 4, day of the year, longitude, and latitude are also used to estimate the vertical distribution of b_{bp} and Chl. Light variables, on the other hand, rely on day of the year and local time.

Table 3: Data sources used to train SOCA2024 and SOCA-light vertical distribution estimations

Field	Data Source	Details
Satellite derived reflectances (R_{rs})	Copernicus Marine Service	OCEANCOLOUR_GLO_BGC_L3_MY_009_103 (cmems_obs-oc_glo_bgc-reflectance_my_l3-multi-4km_P1D dataset)
Satellite derived diffuse attenuation coefficient (KD490)	Copernicus Marine Service	OCEANCOLOUR_GLO_BGC_L3_MY_009_103 (cmems_obs-oc_glo_bgc-transp_my_l3-multi-4km_P1D dataset)
Photosynthetically Available Radiation (PAR)	GlobColour	Photosynthetically Available Radiation L3 reprocessed daily satellite-based observations (http://hermes.acri.fr)
Sea Level Anomaly (SLA)	Copernicus Marine Service	SEALEVEL_GLO_PHY_L4_MY_008_047 (cmems_obs-sl_glo_phy-ssh_my_allsat-l4-duacs-0.25deg_P1D dataset)
Mixed Layer depth (MLD) & Temperature/salinity profiles	Argo	ftp.ifremer.fr/ifremer/argo/; http://www.argo.ucsd.edu
b_{bp}, Chl, PAR, ED380, ED412 and ED490 profiles	Argo	ftp.ifremer.fr/ifremer/argo/; http://www.argo.ucsd.edu

Table 4: Data sources used as inputs to reconstruct MULTIOBS_GLO_BIO_BGC_3D_REP_015_010 Chl, b_{bp} , PAR, ED380, ED412 and ED490 vertical distributions.

Field	Data Source	Details
Satellite derived reflectances (R_{rs})	Copernicus Marine Service	OCEANCOLOUR_GLO_BGC_L3_MY_009_103 (cmems_obs-oc_glo_bgc-reflectance_my_l3-multi-4km_P1D dataset)
Satellite derived diffuse attenuation coefficient (KD490)	Copernicus Marine Service	OCEANCOLOUR_GLO_BGC_L3_MY_009_103 (cmems_obs-oc_glo_bgc-transp_my_l3-multi-4km_P1D dataset)
Photosynthetically Available Radiation (PAR)	GlobColour	Photosynthetically Available Radiation L3 reprocessed daily satellite-based observations (http://hermes.acri.fr)
Sea Level Anomaly (SLA)	Copernicus Marine Service	SEALEVEL_GLO_PHY_L4_MY_008_047 (cmems_obs-sl_glo_phy-ssh_my_allsat-l4-duacs-0.25deg_P1D dataset)
Mixed Layer depth (MLD) & Temperature/salinity vertical distribution	Copernicus Marine Service	MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012 (dataset-armor-3d-rep-weekly dataset)

II.2 Description of the system

This section provides a comprehensive technical overview of the system used for deriving the biogeochemical parameters. It details the core methodologies, including the SOCA2024 method and the development of the SOCA-light-2025 models, the specific databases used for training and validation, the architecture of the neural networks, and the procedures for data conversion and the eventual 3D field reconstruction of the key variables.

II.2.1 General description of the SOCA2024 method

(Sauzède et al., 2016) proposed a neural network-based method (SOCA2016) that merges satellite ocean colour bio-optical products with Argo temperature-salinity profiles to infer the vertical distribution of bio-optical properties (i.e., b_{bp}) in the productive layer. This neural network is a **multi-layer perceptron (MLP)** model trained and validated using a database including 4,725 simultaneous profiles of temperature, salinity and bio-optical properties collected by **BGC-Argo floats** associated with **satellite derived products**. This neural-network based method used **three main input components**:

1) Satellite-based surface estimates of b_{bp} and Chl matched up in space and time,

2) Depth-resolved physical properties derived from temperature-salinity profiles measured by Argo profiling floats (i.e., 4 density values over the euphotic layer and the MLD),

3) Day of the year of the considered satellite-Argo matchup.

SOCA2016 has been upgraded to SOCA2024, the new method to obtain the current high-level 3D gridded **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product by:

- Using a larger amount of new BGC-Argo data since 2016 (more than ten times more data than for SOCA2016 training with less geographical bias of sampling, see Figure 3).

- Using more accurate b_{bp} profiles thanks to correction for slight bias in the b_{bp} sensor calibration factors as identified in (Poteau et al., 2017) and subsequently reprocessing the whole database according to the Argo recommendations (Schmechtig et al., 2018).
- Training a specific neural network to retrieve the vertical distribution of Chl similarly to b_{bp}. The Chl data used for training has been refined following a delayed-mode procedure proposed by the BGC-Argo Data Management Team for improved accuracy. More details about this processing are provided in II.2.
- Using additional inputs such as Sea Level Anomaly (SLA) and Absolute Dynamic Topography (ADT). SLA represents the variable part of sea surface height, providing potential insights into mesoscale processes that can impact the vertical distribution of phytoplankton biomass (McGillicuddy, 2016). ADT is the sum of the variable part (SLA) and the constant part due to the stationary component of ocean circulation, offering information about large-scale general ocean dynamics. These two types of data, although related, provide different insights into ocean dynamics, particularly in the major currents.
- Replacing satellite ocean colour bio-optical inputs Chl and b_{bp} used in SOCA2016 with remote sensing reflectance (R_{rs}) at five wavelengths (i.e. 412, 443, 490, 555, and 670 nm) to avoid inherent errors associated with the use of ocean colour algorithms. The R_{rs} data used as inputs are the median of R_{rs} available for the considered matchup (window of +/- 4 days and 2 bins around the float location, i.e. 8 days and a micropixel of 5x5 pixels).
- Adding the standard deviation of R_{rs} at 412 nm to account for the variability of the pixels used to compute the median R_{rs}.
- Improving temperature-salinity representation in the inputs by replacing the four density values within the euphotic layer used in SOCA2016. Instead, the first Principal Component Analysis (PCA) of the full temperature and salinity profiles was applied.
- Improving the vertical resolution of the outputs (b_{bp} retrieval) in order to obtain b_{bp} at the 36 depth levels from surface down to 1000 m for **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product (determined by the 36 depth levels of the Copernicus MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012 physical product used as input), instead of the 10 previous depths of SOCA2016 solely focusing on the productive layer. The estimation of b_{bp} at higher resolution along the vertical dimension and up to 1000 m depth is of great interest for future carbon export applications.
- Improving the performance of the neural network by using an ensemble of the best Multi-Layer Perceptrons[3,4] (MLPs) (with different topologies, i.e., number of neurons in each hidden layer). Using an ensemble of MLPs also allows us to obtain uncertainties associated to neural network estimations.

[3] https://en.wikipedia.org/wiki/Multilayer_perceptron
[4] https://www.datacamp.com/tutorial/multilayer-perceptrons-in-machine-learning

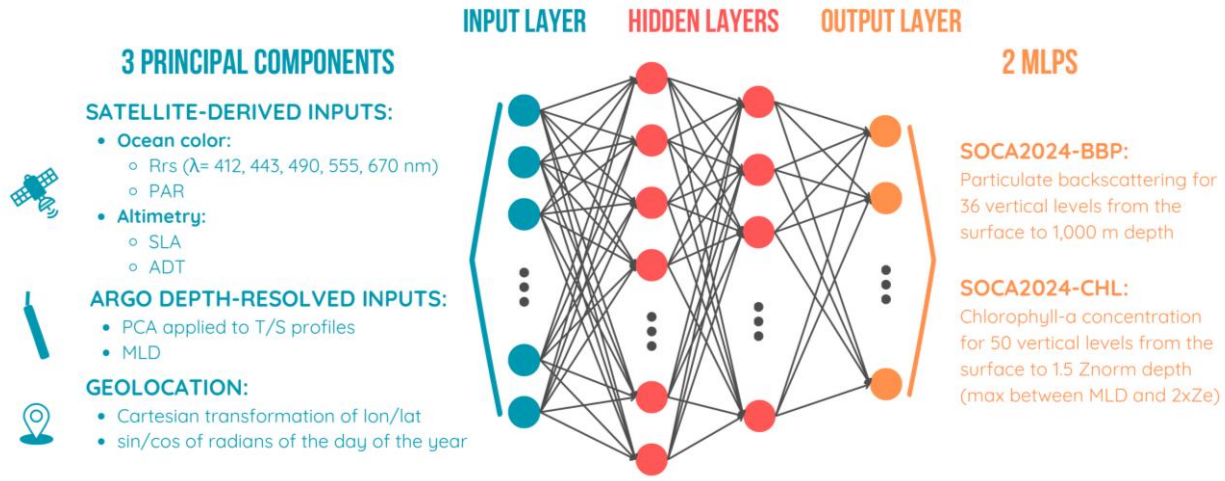


Figure 1: Schematic representation of the SOCA2024 methodology: Two different multi-layer perceptrons retrieve the vertical distribution of b_{bp} or Chl from merged satellite and Argo data, incorporating the longitude, latitude and day of the year of the considered satellite-to-Argo matchup. The depth-resolved vertical profiles of Chl for weekly fields and monthly climatology of the MULTIOBS_GLO_BIO_BGC_3D_REP_015_010 product was generated by linear interpolation from the 50 levels to the 36 vertical depth levels (the same levels as for b_{bp} and POC).

The SOCA2024 neural networks, one designed to estimate b_{bp} and the other to estimate Chl, comprise four layers each: an input layer, two hidden layers and an output layer (Figure 1).

The input layer consists of three main components:

1) **Surface component:** Satellite based surface estimations of remote sensing reflectance (R_{rs}) at 5 different wavelengths (412, 443, 490, 555 and 670 nm), Photosynthetically Available Radiation (PAR), Seal Level Anomaly (SLA) and Absolute Dynamic Topography (ADT).
2) **Vertical component:** Depth-resolved physical properties derived from temperature and salinity profiles measured from BGC-Argo floats (from PCA).
3) **Spatio-temporal component:** the cycle transformation of the day of the year and Cartesian transformation of latitude and longitude. The day of the year is a cyclic variable, so it was converted into radians, the corresponding sinus and cosines were used as inputs to better capture the cyclic feature (see details in Sauzède et al. 2016). Geographic coordinates (longitude, latitude) are converted into Cartesian (x,y,z) coordinates to accommodate Euclidean-centric distance calculations due to the Earth's curvature.

The output layer consists of the b_{bp} and Chl concentrations at different depths

- For b_{bp}, the output layer consists of b_{bp} values at 36 vertical levels from the surface to 1000 m depth.
- For Chl, the output layer consists of Chl concentrations at 50 vertical levels from the surface to 1.5 times a normalization depth depth, ZNORM (see a schematic representation of upgraded SOCA2024 methods in Figure 1. ZNORM is the maximum depth between the MLD and 2 *euphotic depth (Ze). The Chl concentrations are then interpolated to match the 36 vertical levels of b_{bp}. Chl values for depths greater than 1.5 times ZNORM are flagged as *NaN* values.

The robustness and reliability of a neural network can often be significantly improved by combining several neural networks into an ensemble model (Linares-Rodriguez et al., 2013). The construction of a MLP ensemble for both upgraded SOCA2024 neural networks involves two main steps. Firstly, individual model members are created with different architecture (i.e., number of neurons in each hidden layer). Secondly, the outputs of these members are averaged to produce a unique ensemble output.

The selection of the number of MLPs for the ensemble is determined objectively based on a statistical criterion adapted from (Mograne et al., 2019). This criterion defines a scoring scheme that summarizes multiple statistics into a single metric, facilitating the comparison of different models. The score is computed based on the variation range between the minimum and maximum values of each statistical parameter score among all MLPs. The equation of the total score calculation, as presented by (Mograne et al., 2019), was adapted to include the R^2, slope, Median Absolute Percent Difference (MAPD), Root Mean Squared Difference (RMSD). A higher score indicates a better-performing MLP.

To estimate the best ensemble, we first calculate the score of the best single MLPs, followed by the score of the average of the two best MLPs, and continue incrementally including more and more MLPs. The point at which the score begins to decrease with the number of MLPs averaged indicates the optimal ensemble size. Finally, based on this objective scoring method, the optimal ensemble size was determined to be 32 MLPs for Chl and 10 MLPs for b_{bp}. As shown in Table 6 in Section II.2.2, the statistics for the retrieval using the ensemble of MLPs are improved. Hereafter, SOCA2024 refers to the two ensembles of MLPs so derived.

In the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product, the estimation error is available together with each retrieval (i.e., b_{bp}, POC, and Chl). Their errors of b_{bp} and Chl are estimated through the dispersion of the ensemble of neural networks that compose SOCA2024 for b_{bp} (and Chl) retrieval. POC retrieval error is estimated based on the propagation of b_{bp} SOCA-based retrieval error through the b_{bp}-POC relationship, with the additional fixed 20% error included to account for uncertainties in this conversion, as described by Johnson et al. (2017).

II.2.2 Database used for the SOCA2024 training

Argo data processing:

For this study, temperature and salinity data (T and S, respectively) were used in all modes available (i.e., R for real-time, A for adjusted in real-time and D for delayed Mode, (Bittig et al., 2019)) with the available higher accuracy data selected and with an exclusion criteria applied to data flagged with quality control (QC) flags of 3 or 4 (see Reference Table 2 in Argo User's manual[5]).

Bio-optical measurements are derived in their biogeochemical counterparts following the standard Argo protocols: the chlorophyll-a concentration (Chl) is derived from the fluorescence of chlorophyll-a measured by the fluorometer and the b_{bp} is derived from the Volume Scattering Function (VSF) measured by the backscattering sensor (Schmechtig et al., 2015, 2018).

b_{bp} data are used in all modes available (i.e., R, A or D) with data selected based on higher accuracy availability and with an exclusion criteria applied to data flagged with a quality control level 3 or 4. We used data from all modes due to implementation delays by several Argo Data Assembly Centres (DACs) in making A mode data available with the new Real-Time QC tests developed by Dall'Olmo et al., (2023). To address challenges with neural networks in detecting extreme events, such as b_{bp} spikes, b_{bp} profiles

[5] https://archimer.ifremer.fr/doc/00187/29825/

were filtered to remove spikes, following the approach outlined in previous studies by (Briggs et al., 2011, 2020). Consequently, in the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product the b_{bp} and the associated derived POC correspond to b_{bp} and POC associated to small particles (Briggs et al., 2020).

For Chl profiles, due to implementation problems, very few data are available in D-mode in the Argo datastream. For SOCA2024 development, we have applied a specific post-processing that acts similarly to a D-mode adjustment (see the schematic representation of the complete workflow in Figure 2). This D-mode procedure applies three main corrections:

1) Dark correction is applied to account for sensor background noise, which can result in slightly elevated Chl even at depth where Chl should ideally be zero. The dark correction is performed by calculating the median of the minima of Chl values across all profiles collected throughout the float's lifetime. The median value is then subtracted from all Chl measurements to correct for the sensor's baseline noise;
2) Non-Photochemical Quenching (NPQ) correction addresses the decrease in fluorescence observed under high light conditions, which is not due to an actual decrease in Chl-a but rather to photoprotective mechanisms in phytoplankton. The NPQ correction is done following the methodology presented by (Terrats et al., 2020), adapted by (Xing et al., 2018), based on both Photosynthetically Available Radiation (PAR) and b_{bp} measurements;
3) Physiological correction factor of fluorescence into Chl adjusts the fluorescence signal to accurately reflect Chl concentration by accounting for physiological variations in fluorescence yield. This factor is determined from the method of (Xing et al., 2011), based on the vertical profile of the downwelling irradiance at 490 nm (ED490).

This D-mode procedure has been applied to all the BGC-Argo fleet measuring fluorescence, including floats that are non-equipped with a radiometer. For those floats, radiometric parameters (PAR and ED490) were modelled thanks to the SOCA-light method (Renosh et al., 2023). As with the SOCA methods (SOCA2016 and SOCA2024) presented above, SOCA-light needs a satellite ocean colour matchup with Argo profile to derive PAR and ED490 profiles. The availability of an ocean colour matchup is not an issue for the physiological correction factor as the median of all the slopes computed over the whole float lifetime is used, allowing every float to have a slope estimation in D-mode. However, for the NPQ correction that is applied on a profile-by-profile basis, when the ocean colour matchup is not available the SOCA-light-derived climatological PAR profile was used to evaluate the depth up to which the NPQ should be corrected. This NPQ method correction is more accurate than the Xing et al. (2012) method used currently in A-mode, which only takes into account an estimation of the mixed layer depth. All the QC flags were then processed in accordance with BGC-Argo standard protocols (Schmechtig et al., 2023). We applied an exclusion criterion with a quality control level 3 or 4. This new D-mode Chl database was evaluated against the A-mode version by comparing surface data with satellite ocean colour data. This comparison highlights that the new methodology based on quenching correction from Terrats et al. (2020) significantly improves Chl in surface waters, especially in the clear waters of oligotrophic areas. Additionally, we compared the physiological correction factors determined for the delayed-mode corrected Chl dataset with those presented in (Roesler et al., 2017), and found them to be highly consistent.

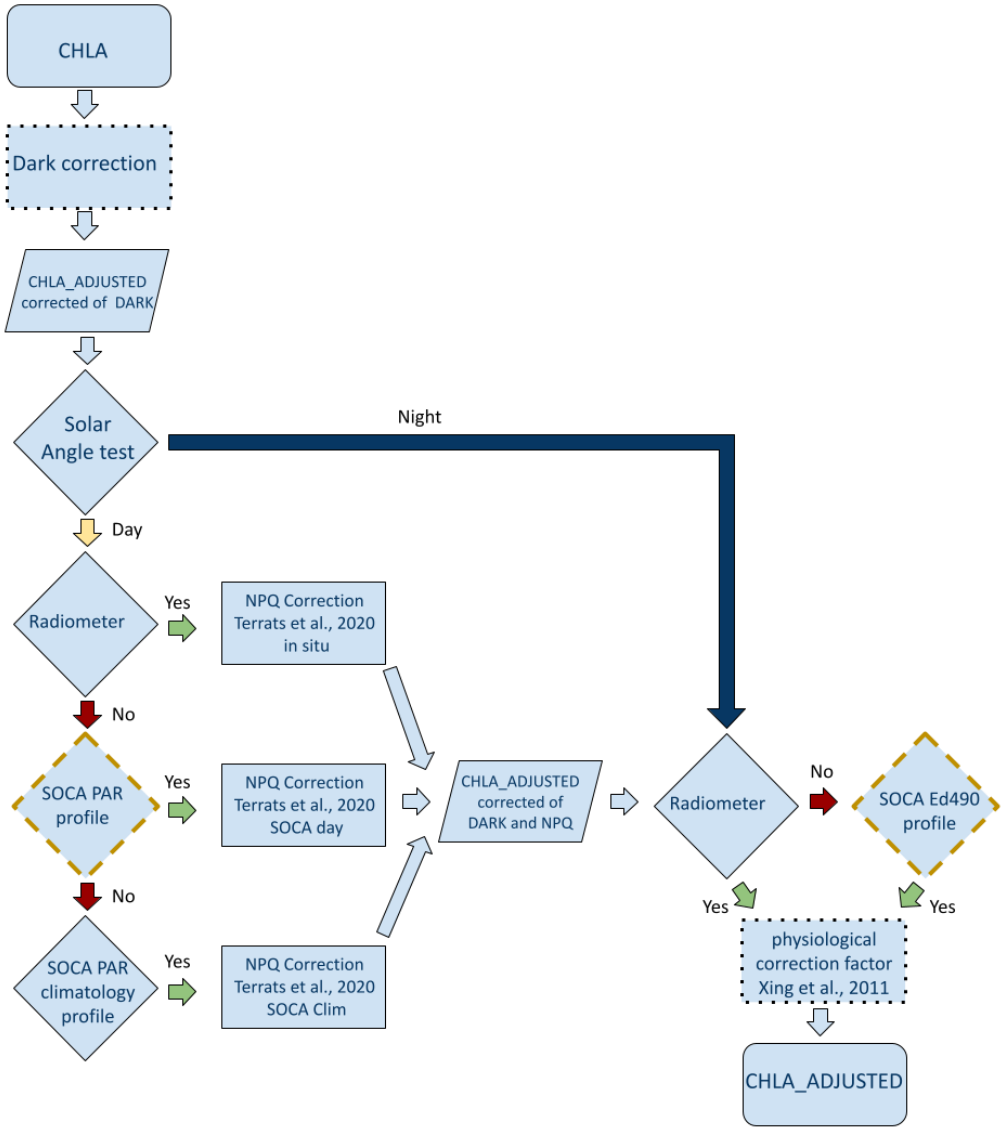


Figure 2: Schematic representation of the post-processing of Chl data in D-mode used for SOCA2024 training. The black dashed rectangle corresponds to the correction applied using the median of data for the entire lifetime of float estimations. The yellow dashed diamonds indicate where a direct ocean colour matchup is required to obtain a value.

Spikes—sharp increases in signal—in the remaining D-mode Chl profiles, mainly due to the fluorescence by large size Chl-containing particles, e.g., phytoplankton aggregates, have been removed using Briggs et al. (2020) methodology. Similarly, for b_{bp}, it is done using a running minimum filter followed by a running maximum filter. Finally, the profiles acquired in the subtropical gyres and Oxygen Minimum Zones (OMZ) —the zone in which oxygen saturation in seawater in the ocean is at its lowest— have been corrected because in these specific areas of known increased background fluorescence signal with depth, due to the increase of Colored Dissolved Organic Matter (CDOM), is known. To correct the artifact fluorescence, increase at depth, the method from (Wojtasiewicz et al., 2018) has been used and adapted to be sure to provide a good correction in case of a second Chl maximum associated with subduction processes, such as described in (Llort et al., 2018). The subtropical gyres and OMZs have been

characterized using climatological composites of surface Chl, computed from satellite data available in CMEMS (OCEANCOLOUR_GLO_BGC_L3_MY_009_103 product) and from World Ocean Atlas of dissolved oxygen at 400 m depth (WOA; (Garcia He et al., 2019). If the surface Chl is lower than 0.08 mg m^{-3} or if the WOA dissolved oxygen is lower than 75 µmol kg^{-1} at 400 m depth for the location of the BGC-Argo Chl profile, the profile is considered to be in a subtropical gyre or in an OMZ, respectively, and will be corrected for Chl at depth.

This resulting Chl and b_{bp} BGC-Argo dataset with application of QC criteria contains 97,985 profiles. After applying the data selection workflow of this BGC-Argo dataset (see details of the selection processes in Table 5), a total of 58,386 simultaneous profiles of temperature-salinity and bio-optical properties, along with corresponding satellite-derived products, are available for developing SOCA2024. In the data selection workflow, in addition to the satellite data matchup and the application of the bathymetric mask, we removed b_{bp} profiles not yet flagged as bad in the BGC-Argo database thanks to a comparison between b_{bp} measured from BGC-Argo floats and the SOCA2020-derived b_{bp} climatological counterpart (see details in the b_{bp} audit provided by the BGC-Argo Data Management Team[6]). Moreover, using the method of (Haëntjens et al., 2020), all b_{bp} profiles with a zooplankton signature have been removed from the training database. In fact, traditionally POC refers to phytoplankton and non-algal particles in the same range size, thus excluding zooplankton. The corresponding Chl profiles have been removed in order to have a consistent database to be used to develop both MLPs for the b_{bp} and Chl retrieval. The geographical distribution of the 58,386 profiles used for developing SOCA2024 is shown in Figure 3.

SOCA2016 database was biased as it was characterized by an under sampling of the southern hemisphere for the training dataset. The southern hemisphere representation in the SOCA2024 database is improved thanks to the SOCCOM program that deployed BGC-Argo profiling floats in the Southern Ocean since 2015 (right panel in Figure 4). On a monthly basis, more profiles are available for spring and summer than for autumn and winter months for the northern and southern hemisphere (left panel in Figure 4). This temporal bias of data acquisition is mostly due to a lack of satellite ocean images at high latitudes during winter and autumn as well as to an improved temporal resolution of float acquisition in spring and early summer to catch key biogeochemical processes like blooms and export events. Although the annual distribution of the vertical profiles in the database covers 14 years from 2010 to 2024 (left panel in Figure 4), most of the observations are collected since 2013.

Table 5: Description of the number of b_{bp} and Chl profiles available to develop SOCA2024 after each data selection step (continue in the next page).

Data selection steps	Available profiles	Number of removed profiles (% of data removed for each step)
Total b_{bp} and Chl profiles available		-
P/T/S profiles flagged as bad	113 945	6 325 (~ 5%)
b_{bp} profiles flagged as bad	99 778	14 167 (~ 12%)
Chl profiles flagged as bad	97 985	1 793 (~ 2%)
Black Sea mask	96 854	1 131 (~1%)
bathymetric mask (1500 m depth)	93 254	3 600 (~ 4%)
"Full" profiles (several measurements available across depth layers: at least one in the 0-25 m layer, three in each of the 25-250 m and 250-1000 m layers, and at least one below 900 m)	79 145	14 109 (~ 15%)

[6] Available ftp://ftp.mbari.org/pub/BGC_argo_audits/BBP700

Table 5: (continued)

Data selection steps	Available profiles	Number of removed profiles (% of data removed for each step)
with satellite altimetry matchup	77 200	1 945 (~ 2%)
with satellite ocean colour matchup	60 958	16 242 (~ 21%)
b_{bp} profiles not yet flagged as bad in the BGC-Argo database	60 243	715 (~1%)
additional data considered bad and rejected for this study (i.e., zooplankton signature for b_{bp})	58 386	1 857 (~ 3%)

Before splitting the BGC-Argo and satellite matchup database into two subsets to develop the neural network (i.e., the so-called training and validation datasets), 908 profiles collected by six floats were removed from the database to create an "independent data set" to use for additional validation of the method. These BGC-Argo time-series were chosen in major oceanic basins: the Subpolar and Subtropical gyres of the North Atlantic, the Atlantic Equatorial Waters, the Western and Eastern Mediterranean Sea and the Southern Ocean (see the coloured points in Figure 3). These floats are referring to the official World Meteorological Organization (WMO) numbers 6901523, 6901472, 4903459, 6901491, 6901773 and 6901493, respectively. The resulting was randomly split into two independent subsets: 45,982 (≈80%) profiles used for training the MLP and 11,496 profiles (≈20%) used for validating the MLP.

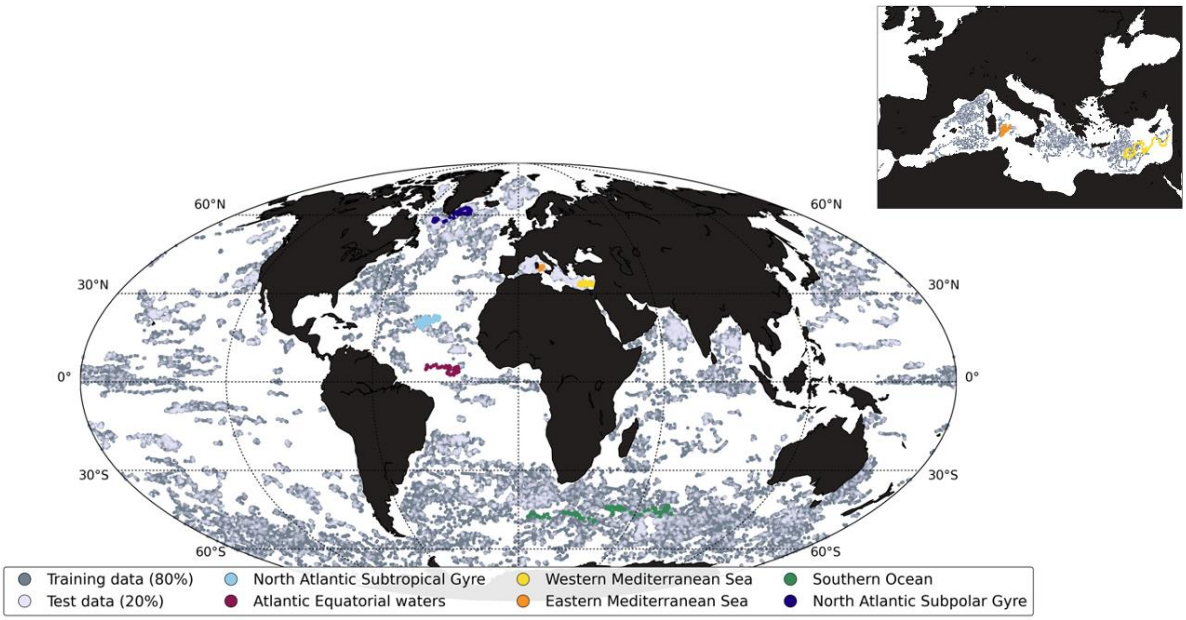


Figure 3: Geographical distribution of the 58,386 stations used for developing SOCA2024. For each station, concurrent profiles of temperature, salinity, Chl and b_{bp} collected by BGC-Argo floats were matched up with concomitant satellite products. Dark grey points: training dataset. Light grey points: validation dataset. Vertical profiles from six independent BGC-Argo floats shown in different colours based on geographical position. Zoom of the Mediterranean Sea in the upper right panel.

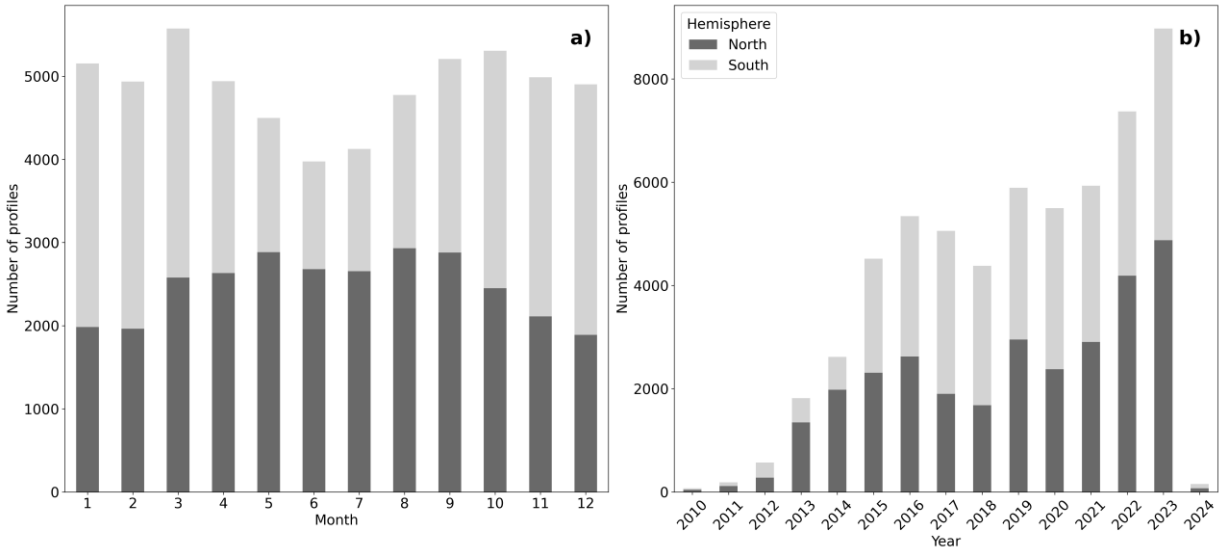


Figure 4: Temporal distribution of the 58,386 stations for which both BGC-Argo Chl and b_{bp} profiles and satellite data were simultaneously available as a function of months (panel a)) and years (panel b)) with dark and light grey colours indicating the hemisphere of data acquisition.

II.2.3 SOCA2024 neural networks details

Inputs:

In order to reduce the dimensionality of the vertical input profiles, we applied a Principal Component Analysis (PCA) to the temperature and salinity (T and S) profiles. To achieve 0.998 of the cumulative variances, in SOCA2024-BBP we selected the first seven principal components for Temperature, and the first six for Salinity. In SOCA2024-CHL, we selected the first five for Temperature, and the first four for Salinity. The difference in the number of principal components selected for both models is due to the different resolutions of the Temperature and Salinity profiles in SOCA2024-BBP (36 vertical levels from the surface to 1,000 m depth) and SOCA2024-CHL (50 vertical levels from the surface to the normalization depth).

Integration of a normalization depth for Chl retrieval:

In the same way as in (Sauzède et al., 2015), we integrated a normalization depth (ZNORM) to better account for the vertical variability of Chl profiles in contrasting open ocean environments. This includes typical profiles of deep winter mixing conditions, those characterized by a Deep Chl Maximum (DCM) in stratified oligotrophic systems, and profiles with a subsurface Chl maximum found in mesotrophic or eutrophic environments with relatively shallow mixed layer depths (MLD).

Chl is retrieved for 50 vertical levels up to ZNORM. To obtain Chl values for the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product, we applied a linear interpolation to the 50 output depths, providing values at the 36 vertical depth levels, consistent with those used for b_{bp}. The Chl values for depths greater than 1.5 times ZNORM are flagged as *NaN*.

MLPs architecture and ensemble:

After multiple tests, the MLPs architecture selected as optimal is composed of 28 and 24 inputs for Chl and b_{bp}, respectively. Both MLPs contain nine surface inputs, five spatio-temporal inputs. Moreover SOCA2024-BBP contains 14 vertical inputs and SOCA2024-CHL contains 10 vertical inputs. The two SOCA2024 MLPs contain 36 and 50 outputs for b_{bp} and Chl, respectively. The SOCA2024 method is composed of an ensemble of 10 MLPs for b_{bp} and 34 for Chl. As mentioned above, the statistics for the retrieval using the ensemble of MLPs are improved when compared to the one MLP with the architecture of best retrieval statistics (Table 6).

Table 6: Statistics of the comparison between b_{bp} and Chl values predicted by the neural networks (MLP) and BGC-Argo reference measurements (r^2: regression coefficient; RMSD: Root Mean Square Difference; MAPD: Mean Absolute Percentage Deviation).

Metrics	MLP for b_{bp} retrieval (architecture with the best statistical performance)	SOCA2024 for b_{bp} retrieval: MLP ensemble	MLP for Chl retrieval (architecture with the best statistical performance)	SOCA2024 for Chl retrieval: MLP ensemble
r^2	0.89	0.90	0.82	0.87
Slope	0.90	0.90	0.83	0.93
MAPD	12 %	11 %	32%	29%
RMSD	0.00029 m-1	0.00025 m-1	0.20 mg m-3	0.19 mg m-3
Bias	-0.00002 m-1	0.00002 m-1	0.02 mg m-3	0.001 mg m-3

II.2.4 Conversion of b_{bp} into Particulate Organic Carbon

A dedicated POC *vs.* b_{bp} at 700 nm relationship for the global ocean has been specifically developed and used from the second release of **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product (the one developed by (Cetinić et al., 2012) was used for the previous releases). The *in situ* POC database matched up with satellite data from (Evers-King et al., 2017), which gathers ~6,500 samples for the global ocean (see Figure 5), has been used to establish the POC *vs.* b_{bp} relationship. We have refined this relationship for this version of the product. First, for each station with POC samples, the SOCA2024-b_{bp} value at 700 nm, the reference wavelength for BGC-Argo measurements, has been derived for the corresponding depth of POC sampling. Then, different relationships have been tested (i.e., simple linear relationship, multi-linear relationship, and exponential relationship). The best relationship that was found is a power law relationship (see Figure 6):

$$POC = 66179.11 \times {b_{bp}}^{1.022} \quad (1)$$

For comparison, the relationships of (Cetinić et al., 2012), (Stramski et al., 2008), (Balch et al., 2010), and (Koestner et al., 2024) are shown in Figure 6. Based on performance retrieval statistical (data not shown), the relationship developed in this study is the most suitable, even when compared to the recently published relationship by (Koestner et al., 2024). However, if users prefer to apply other existing relationships to derive POC from b_{bp}, they can easily do so, as b_{bp} is readily available in the product.

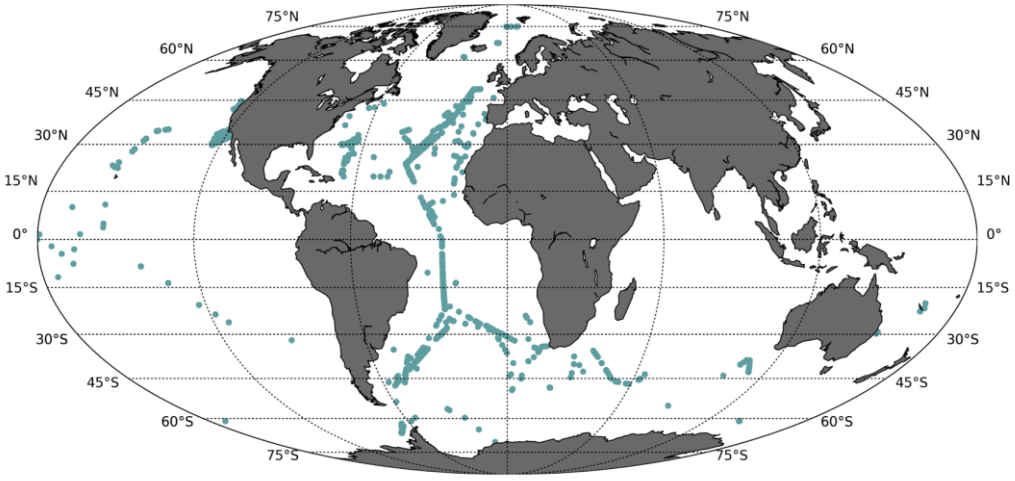


Figure 5: Geographical distribution of the ~6,500 samples from the database published by Evers-King et al. (2017) that was used for developing the POC vs. $b_{bp}(700)$ relationship used in the MULTIOBS_GLO_BIO_BGC_3D_REP_015_010 product.

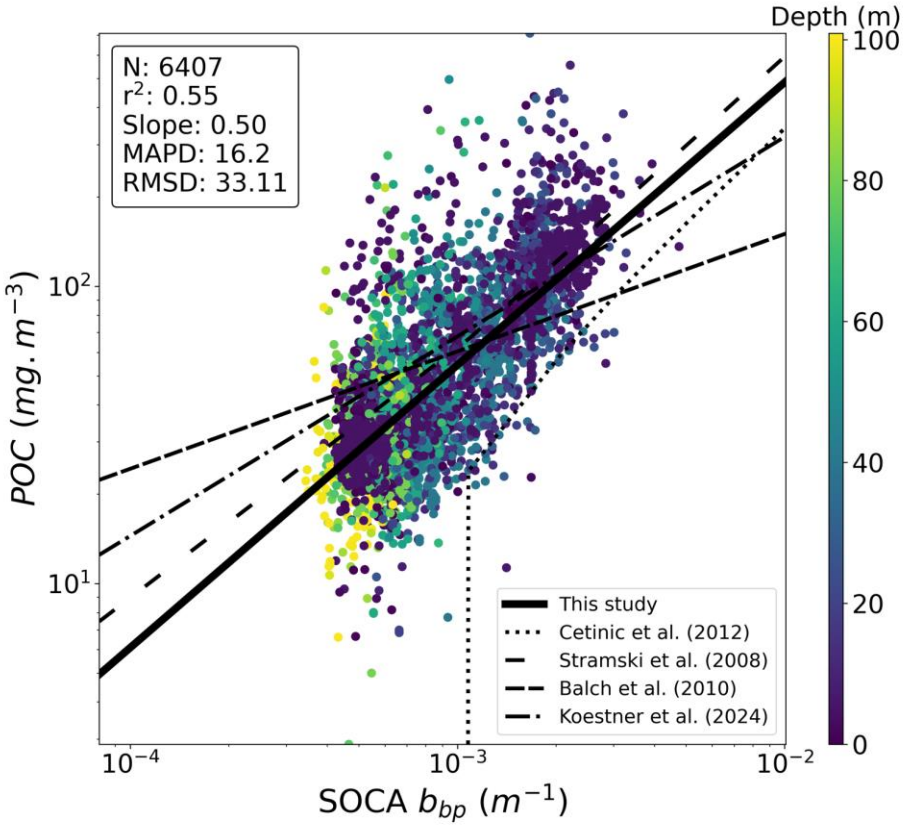


Figure 6: Scatterplot between in situ POC and SOCA2024-derived $b_{bp}(700)$ with colour referring to the depth of POC sampling and lines referring to different relationships (see legend).

The power law relationship (1) retrieves POC for the global ocean and for all seasons with a Median Absolute Percent Difference (MAPD) of 16% and a Root Mean Squared Difference (RMSD) of 33 mg m^{-3}. By comparison, the relationship by (Cetinić et al., 2012) applied to this global database retrieves the POC with a MAPD of 75% and a RMSD of 51 mg m^{-3}, which means that (1) represents a significant improvement from the second release of the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product. Note that for the fourth release of the product, the relationship has been updated using the new SOCA-2024 b_{bp} (700) values.

II.2.5 General description of SOCA-light-2025 models

Expanding on the original SOCA models, that successfully estimated bio-optical properties like Chl and b_{bp} from BGC-Argo float observations, we have developed the SOCA-light models a refined extension specifically designed to generate vertical profiles of underwater light, including PAR, ED380, ED412, and ED490, in open-ocean environments (Renosh et al., 2023). These models are powered by a unique dataset spanning over ten years, comprising extensive light measurements collected by BGC-Argo floats (Claustre et al., 2020). They utilize the SOCA methodology, which integrates satellite-derived surface ocean colour data with *in situ* measurements of temperature and salinity collected by BGC-Argo floats. At the core of the approach is an artificial neural network (ANN) trained on a unique dataset of light profiles specifically PAR and ED obtained from the BGC-Argo floats. The model inputs include satellite ocean colour data and Argo-based temperature and salinity profiles, while the outputs are depth-resolved PAR and ED profiles. Through rigorous validation using independent datasets, the models have demonstrated high consistency and accuracy, underscoring their robustness and suitability for a wide range of oceanographic applications. These models mark a significant advancement in our ability to assess underwater light conditions, which are crucial for photosynthesis, marine primary productivity, and the behaviour of light-sensitive organisms. They overcome longstanding challenges related to the sparse spatial and temporal resolution of direct underwater light measurements in marine environments. In 2025, the SOCA-light model was upgraded through retraining, incorporating a larger number of profiles (Table 7) and adjustments to the input variables (SOCA-light-2025).

II.2.6 Database used for the SOCA-light-2025 model training

BGC-Argo data: BGC-Argo floats (Claustre et al., 2020) equipped with multi-spectral ocean colour radiometers (Satlantic OCR-504, Satlantic Inc., Halifax, NS, Canada) measuring ED at three different wavelengths, i.e., 380, 412 and 490 nm, W $m^{-2}nm^{-1}$, and PAR, µmol quanta m^{-2} s^{-1}, were used for the present study. From among the synthetic BGC-Argo individual profiles available at the Coriolis Global Data Assembly Center (GDAC) (Bittig et al., 2022), only radiometric measurements qualified in delayed-mode (DM) using the quality control and calibration procedures proposed by Jutard et al., 2021 were kept for the model development. These procedures identify and correct radiometric profiles for any sensor drift or temperature dependence. The correction relies on the acquisition of at least one night profile per year (for the assessment of sensor temperature dependence) and daily dark measurements when the float drifts at the 1,000 dbar parking depth (for the assessment of sensor drift). The previous SOCA-light model estimated surface (0-) ED or PAR values using only a simple linear interpolation, as direct radiometric measurements are unavailable. The updated SOCA-light-2025 significantly improves upon this methodology. It first integrates estimates derived from four distinct interpolation schemes: linear, log-linear, log-linear fitting up to the ZPD (calculated using a satellite matchup of KD490), and a log-nonlinear (second-degree polynomial) fitting up to the ZPD. From this set of comprehensive

estimates, the final surface value (0-) is computed as the median of all interpolated values that exceed the initial ED or PAR measurement, resulting in a much more robust surface estimation. Concurrently with radiometric profiles, DM-qualified profiles of pressure (P), temperature (T), and salinity (S), were also used for the present study. The P, T, and S profiles with a number of qualified measurements less than 5 in the upper 50 m and less than 15 in the upper 250 m were discarded from the present analysis. The geographical locations of all profiles (P, T, S, and PAR) used for the development and validation of the SOCA-light-2025 model for PAR are shown in Figure 7.

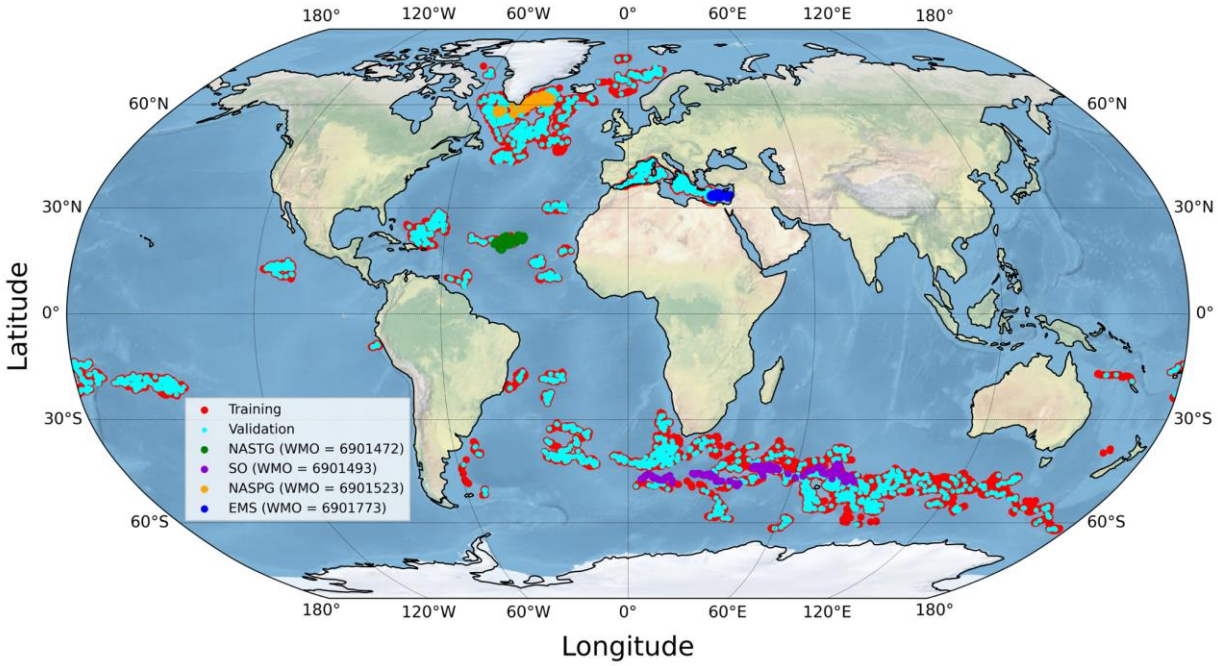


Figure 7: Geographical distribution of BGC-Argo profiles (16,496 profiles) used for the development and validation of the SOCA-light-2025 model for photosynthetically available radiation (PAR) profiles.

Table 7: Number of profiles used for the development SOCA-light and SOCA-light-2025 models.

Variables	Database	SOCA-light	SOCA-light 2025
PAR	Train dataset (80%)	6,636	**12,576**
	Validation dataset (20%)	1,660	**3,144**
	Independent dataset	536	**776**
ED380	Train dataset (80%)	6,641	**12,682**
	Validation dataset (20%)	1,661	**3,171**
	Independent dataset	536	**782**
ED412	Train dataset (80%)	6,642	**12,672**
	Validation dataset (20%)	1,661	**3,168**
	Independent dataset	536	**781**
ED490	Train dataset (80%)	6,642	**12,676**
	Validation dataset (20%)	1,661	**3,170**
	Independent dataset	536	**780**

Satellite Ocean Colour data: For the neural network development and validation, we used satellite-based level-3 (L3) ocean colour products of fully normalized remote sensing reflectance (R_{rs}), PAR, and KD490 from GlobColour products. While (R_{rs} and KD490) data were available from the Copernicus-GlobColour product (**OCEANCOLOUR_GLO_BGC_L3_MY_009_103**), PAR (not similarly available) was directly downloaded from the GlobColour website (http://hermes.acri.fr, accessed on 14 March 2025). These global L3 products, which have a spatial resolution of 4 km, correspond to daily composites obtained from merged L3 Ocean Colour outputs from different sensors, which ensures data continuity, improves spatial and temporal coverage, and reduces data noise (Garnesson et al., 2019).

Selection of the SOCA-light-2025 Training Database: BGC-Argo profiles, together with satellite products measuring ocean colour, made up the initial database for neural network training and validation. The ocean colour matchup was built by selecting the nearest available measurement both in time (within ±5 days) and space (within a 5 × 5-pixel area) relative to the float location and sampling time. Based on the monthly distribution of light profile acquisitions (Figure 8A), it appears that this database does not present any temporal bias in terms of the number of profiles per month globally. However, a seasonal geographical bias exists as fewer profiles exist for the northern and southern hemispheres during their respective winter months. This is due to the reduced number of matchups available because of increased cloud coverage during the winter. The present study utilizes all profiles obtained under conditions where the solar elevation exceeds 5 degrees. On an hourly basis, 97% of the profiles were sampled between 10 and 13 local hours (Figure 8B).

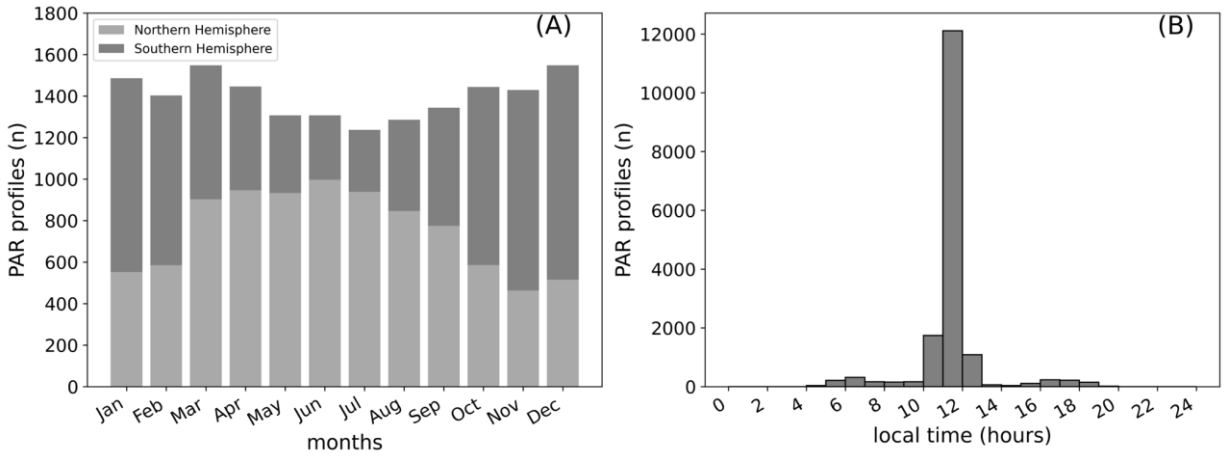


Figure 8: Temporal distribution -monthly (A) and hourly (B)- of PAR profiles used for this study.

From this initial database, separate databases were created for each of the four models (i.e., PAR, ED380, ED412, and ED490). The databases thus constituted were used for SOCA-light-2025 development, with 80% of profiles being used for model training and the remaining 20% for model validation. These training and validation databases were randomly selected. In parallel, the data from four floats with World Meteorological Organization (WMO) numbers 6901472, 6901493, 6901523, and 6901773 were kept aside for independent validation, i.e., these floats were not part of the training and validation processes. These four floats acquired multi-year measurements in four different oceanic areas considered to cover a large range of hydrological and bio- optical conditions typically representative of open ocean waters.

II.2.7 SOCA-light-2025 details

The SOCA-light models, derived from the generic SOCA methodology (Sauzède et al., 2016; Renosh et al., 2023), comprise four distinct models. These models predict vertical profiles of PAR, ED380, ED412, and ED490 at specific geographic locations. They utilize matchup data with satellite ocean colour products and vertical profiles of temperature and salinity as inputs. For SOCA-light-2025, we adapted the input variables used in other SOCA models (i.e., for Chl and b_{bp}) by selecting key variables that represent the vertical propagation of light in the water column, specifically the first optical depth (Z_{pd}). Consequently, unlike other SOCA models that use sea-level anomaly (SLA) to infer mesoscale processes affecting phytoplankton biomass, SOCA-light models exclude SLA from their key variables. In SOCA-light-2025, the input configuration was modified by removing local time and introducing solar elevation as a new input variable. The four SOCA-light-2025 neural networks were trained using a database of concurrent profiles of temperature, salinity, and the light variables (PAR, ED380, ED412, and ED490) collected by BGC-Argo floats and collocated with satellite-derived products. A schematic representation of all SOCA-light-2025 models is shown in Figure 9.

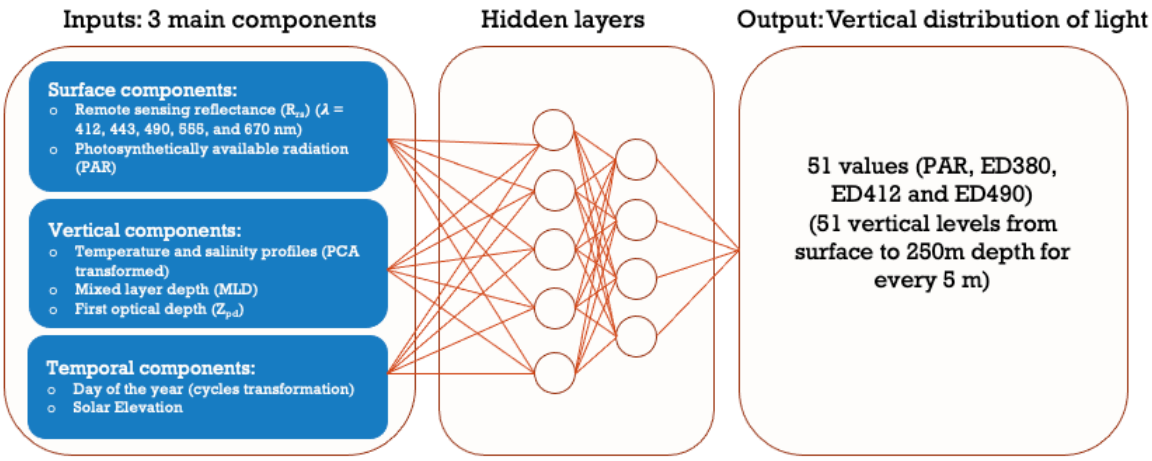


Figure 9: Schematic representation of the SOCA-light-2025 multilayer perceptron. See the main text below for a description of the components.

There are three main input components used for this model (Figure 9):

• **Surface components:** These encompass satellite-based surface estimates of R_{rs} at five different wavelengths (i.e., 412, 443, 490, 555, and 670 nm) and PAR.

• **Vertical components:** These rely on the first principal component analysis of salinity and temperature profiles. The principal components were selected on the basis of cumulative explained variance values less than or equal to 0.998. For temperature, this criterion is satisfied by five principal components, and for salinity, by four principal components. The mixed layer depth (MLD) was derived from density calculated from pressure, temperature, and salinity profiles with a density differential threshold criterion of 0.03 kg m^{-3} with reference to the density at 10 m (de Boyer Montégut et al., 2004). The Z_{pd} was derived from the satellite-derived KD490 using Equation (2).

$$Z_{pd} = \frac{1}{KD490} \quad (2)$$

• **Temporal components:** The temporal components are the day of the year (DOY) and the solar elevation of the sampling profile. These components follow periodic evolution within certain time windows (0 to 365 days for DOY). The cyclic transformations (sine and cosine) of radian-transformed DOY were used as temporal components Equation (3) along with solar elevation:

$$DOY_{rad} = \frac{DOY \times \pi}{182.625} \quad (3)$$

The SOCA-light-2025 model outputs are the four light variables (PAR, ED380, ED412, and ED490) at 51 vertical levels from the surface to 250 m depth at every 5 m interval. The use of an ensemble of MLPs proved effective in improving the robustness and reliability of predictions compared to the use of a single MLP (Linares-Rodriguez et al., 2013). For this reason, as a first step several MLPs were created, each with a unique architecture incorporating the hyperbolic tangent (tanh) as the activation function and adaptive moment estimation (ADAM) as the solver. The ADAM solver streamlines the conclusion of iterations upon reaching model convergence, speeding up the process. At the same time, we identify the optimal number of epochs to ensure effective learning and prevent overfitting. The key distinction among these models lies in the varying number of neurons distributed across each hidden layer, with the intention of capturing diverse patterns and representations inherent in the data. We chose two hidden layers from the considered options of one, two, and three hidden layers for these light models. Notably, models with two hidden layers consistently outperformed, with the number of neurons in the second hidden layer always being fewer than or equal to that in the first hidden layer. The models were trained by changing the neuron numbers between 5 and 150 with an increment of one (altogether 10,585 iterations). The second step was then to select, from all these iterations, an ensemble of the 10 best MLPs based on minimum statistical metrics obtained from training and validation datasets (r^2, RMSD and MAPD). Through this selection, the ensemble model aimed to capture diverse representations while ensuring the sound performance and consistency of individual MLPs.

II.2.8 Reconstruction of 3D fields of b_{bp}, POC, Chl, PAR, ED380, ED412 and ED490

As shown in Table 4, we used reflectance, sea level anomaly, and PAR satellite data as well as temperature-salinity profiles and MLD data available from CMEMS and GlobColour website to estimate the vertical distribution of b_{bp} and Chl, with b_{bp} profiles being subsequently transformed into POC.

For the reconstruction of 3D weekly fields (b_{bp}, POC, and Chl) from SOCA2024, we have used weekly temperature-salinity and MLD fields[7] as inputs and we averaged, on the same period as temperature-salinity and MLD weekly fields (from Wednesday to Wednesday) daily reflectance fields[8], daily SLA fields[9] and daily PAR fields[10]. For the 3D weekly fields of SOCA-light-2025 (PAR, ED380, ED412, and ED490), we included the KD490 daily[11] fields and excluded SLA. Weekly input processing followed the same approach as in SOCA2024. For the reconstruction of 3D multi-year monthly climatological fields (b_{bp}, POC, Chl, PAR, ED380, ED412 and ED490), we averaged the monthly temperature-salinity and MLD fields, the daily reflectance fields, the daily KD490 fields, the daily SLA fields, and the daily PAR fields for every year over the same period as weekly fields. Note that to remove outliers from Arctic Ocean colour

[7] Copernicus MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012 product
[8] Copernicus OCEANCOLOUR_GLO_BGC_L3_MY_009_103 product
[9] Copernicus SEALEVEL_GLO_PHY_L4_REP_OBSERVATIONS_008_047 product
[10] available from GlobColour website: https://hermes.acri.fr/
[11] Copernicus OCEANCOLOUR_GLO_BGC_L3_MY_009_103 product

data (very low values probably due to the ice coverage), we applied a condition that above 70°N a minimum of 60 pixels on the whole time series must be available to compute the climatological satellite data. For both products (i.e., weekly and monthly climatologies), the satellite data were averaged on the same horizontal resolution as the physical fields (i.e., 0.25° x 0.25°). We used PAR data from GlobColour website because no GlobColour PAR data is available from the Copernicus Marine Service portal.

Note that although a bathymetric mask of 1,500 m was applied to the training database to ensure the removal of any ground effects on b_{bp} values at 1,000 m depth, the bathymetric mask applied to create the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product is 1,000 m. The previous 1,500 m mask, used for product versions before November 2024, was found to be too restrictive in regions such as the North Atlantic, which is important for carbon export studies and contains extensive continental shelves. Before deciding to change the bathymetric mask, we verified that the new 1,000 m bathymetric mask did not result in spurious estimations closer to the coast.

Chl is retrieved from the SOCA2024 model, which provides data at 50 vertical levels up to 1.5 times ZNORM depth. To obtain Chl values for the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product, we applied linear interpolation to these 50 output depths, generating values at 36 standard vertical depth levels from the surface down to 1,000 m. These 36 levels are consistent with ARMOR3D data and match those used for b_{bp}. Chl values at depths greater than 1.5 times ZNORM are flagged as *NaN*. Similarly, SOCA-light-2025 provides light variables (PAR, ED380, ED412, and ED490) at 51 vertical depths from the surface down to 250 meters. In the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product, these variables are also linearly interpolated to the same 36 depth levels used for Chl and b_{bp}. Values for the light variables at depths below 250 meters are flagged as *NaN* in this product.

III VALIDATION FRAMEWORK

To assess the performance and reliability of the **SOCA2024** and **SOCA-light-2025** models, a comprehensive validation framework was established. The validation framework for the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product evaluates the SOCA2024 and SOCA-light-2025 model's ability to reproduce the vertical distribution of Chl, b_{bp}, POC, PAR, ED380, ED412 and ED490. This approach ensures a robust and objective assessment of both models, supporting their application in global ocean biogeochemical analyses.

This validation is conducted in several steps:

- First, a global validation (metrics such as Slope, r^2, MAPD, and RMSD) uses 20% of the global dataset that was not included in training. However, as this subset is not entirely independent (since successive profiles may be spatially and temporally close),
- An additional validation step is conducted using entirely independent time series from specific floats that were excluded from both the training and the 20% validation subsets. This added level of validation provides a more rigorous, independent assessment of the model's performance,
- Validation has also been conducted against reference *in situ* datasets of biogeochemical measurements wherever possible to address these specific uncertainties, as transforming bio-optical measurements from BGC-Argo into biogeochemical variables introduces additional uncertainties in the final biogeochemical estimates available in the product.

  For instance, Chl estimates have been cross-validated against Chl values derived from high-performance liquid chromatography (HPLC), providing a trusted benchmark. However, final validation for POC is limited because POC estimation depends on b_{bp}/POC relationships derived from a POC *in situ* dataset specifically gathered for this study, meaning validation using the same data would lack independence. Light variables (PAR, ED380, ED412 and ED490) were cross-validated using the global SeaBASS database, which compiles measurements from a range of cruises and field campaigns. It is worth noting that, unlike BGC-Argo light measurements taken under varying sky conditions, ship-based measurements in SeaBASS are typically collected under clear skies and are more operator-dependent.

In this way, the framework provides a multi-layered assessment of model performance, ensuring reliability and accuracy while acknowledging limitations in cases where independent validation data are not feasible.

IV VALIDATION RESULTS

The accuracy of the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product from January 1998 to December 2019 has been assessed by comparison with *in situ* independent observations. Global accuracy, determined by comparing product values to independent *in situ* observations, shows that Chl has an RMSD of 0.59 mg m^{-3} (MAPD: 47%), while the b_{bp} has an RMSD of 0.00028 m^{-1} (MAPD: 11%). We estimate that, taking into account both the error in b_{bp} retrieval from SOCA2024 and the performance of the b_{bp} sensor in estimating POC from BGC-Argo profiling floats (Johnson et al., 2017), the POC is estimated with an accuracy of approximately 20 mg m^{-3} and a MAPD of around 30%.

Even if no systematic bias was evident from the qualification of the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product, the deep POC and Chl maximums seem to be relatively underestimated. For the retrieval of the deep Chl maximum, although the MULTIOBS_GLO_BIO_BGC_3D_REP_015_010 Chl product appears much more satisfactory than former products resulting from statistical parameterization (e.g., (Uitz et al., 2006)), SOCA2024 still slightly underestimates Chl retrieval at this specific depth.

More details are given in the sections below.

IV.1 Particulate backscattering coefficient (proxy of particulate organic carbon)

The validation for the derived b_{bp}, which serves as a proxy for POC, is presented in this section. The performance assessment is systematically divided into two parts: a large-scale statistical validation against a 20% subset of the BGC-Argo database and a detailed, independent comparison against specific BGC-Argo floats.

IV.1.1 Validation against a 20% random subset of the BGC-Argo database

First, the performance of the neural network-based method is evaluated using the independent validation dataset, i.e. 20% of the initial SOCA2024 database chosen randomly, through the comparison between SOCA2024 b_{bp} retrieval against the BGC-Argo b_{bp} reference measurements (Figure 10 and Table 7). SOCA2024 improves the statistics compared with SOCA2016 and infers the vertical distribution of b_{bp} with a global error of 11% (i.e., median absolute percent difference). SOCA2024 predicts b_{bp} without systematic bias according to the depth of retrieval. Figure 10 shows that most of SOCA2024 b_{bp} values are retrieved with substantial accuracy and that only a very limited number of data points diverge from the 1:1 line.

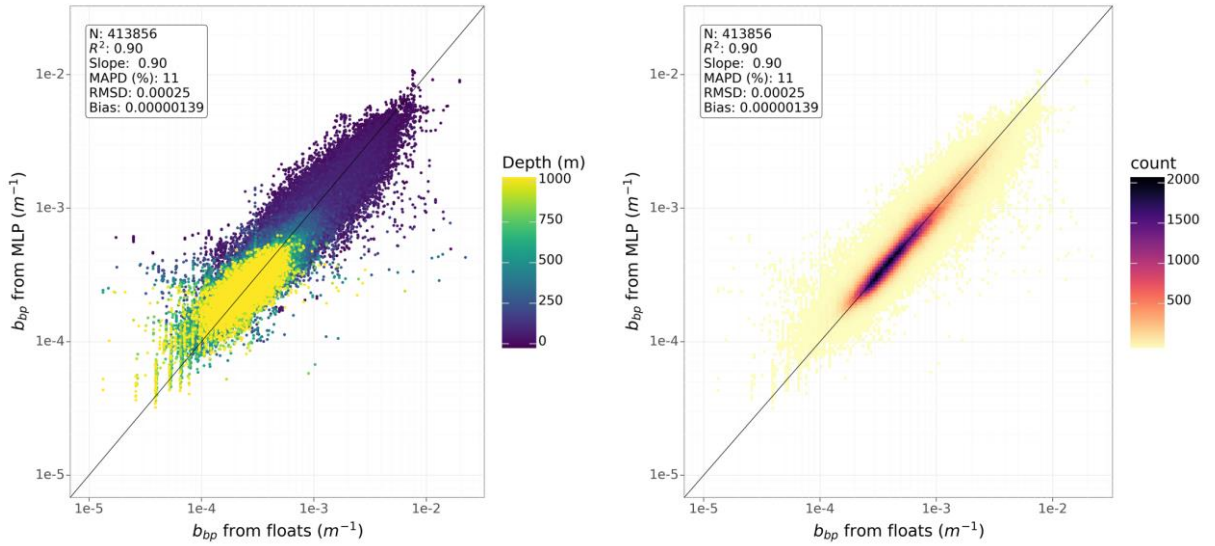


Figure 10: Comparison of the b_{bp} values retrieved by SOCA2024 with reference measurements acquired by BGC-Argo profiling floats from the independent validation dataset, i.e., 11,496 profiles chosen randomly from the entire database The left panel presents a scatterplot with points coloured by depth, while the right panel shows a scatterplot with points coloured according to the density of points in each bin. The 1:1 line is shown in black and the detailed statistics are provided.

Figure 11 shows that b_{bp} is retrieved with slightly less accuracy in high latitude environments, which are the areas where more seasonal and spatial variability in the vertical distribution of b_{bp} can be observed. The phytoplankton has time scales of approximately one day (e.g., biomass can double every day during blooms) while sub mesoscale forcing (few kilometers) can have a significant impact on phytoplankton biomass and subsequently on POC. In this context, it is obvious that the temporal and spatial representability of a matchup (here seven days and pixels of 0.25° x 0.25°) have a large uncertainty. However, it is important to note that no systematic geographical bias can be seen in Figure 11.

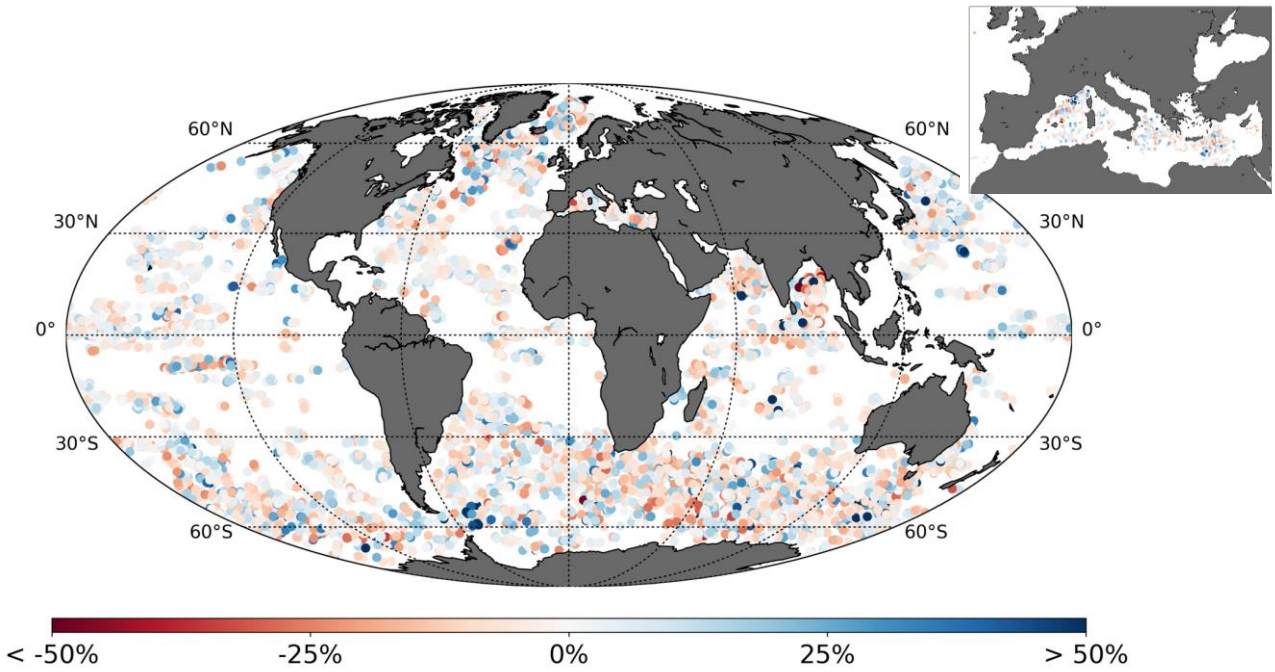


Figure 11: Geographical distribution of the median percent difference for b_{bp} for each profile between SOCA2024-estimates and BGC-Argo data. Zoom of the Mediterranean Sea in the upper right panel.

IV.1.2 Validation against six independent BGC-Argo floats

To further validate and assess the accuracy of the vertical distribution of b_{bp} inferred by SOCA2024, six independent BGC-Argo profiling float time series, previously removed from the database used to train and validate the method, are now used to create a completely independent dataset. The method has slightly better accuracy when tested with this independent data set as with the validation data set comprising 20% of the initial database (Figure 12 and Table 8).

The SOCA2024 time series are compared to their float counterparts across six basins (Figure 13 and Figure 14). Note that the absence of SOCA2024 b_{bp} data (visible as vertical white bands) corresponds to missing satellite-to-Argo matchups, often due to the lack of satellite image in cloudy regions or during winter in high latitudes. The vertical patterns of b_{bp} predicted by SOCA2024 are highly consistent with those observed by profiling floats within the 0-1000 m layer. In the North Atlantic Subtropical Gyre, an area characterized by extremely low b_{bp} values, SOCA2024 successfully reproduces the seasonal deepening of the b_{bp} maximum in spring and early summer, as well as the subsequent shoaling in June–July, throughout nearly six years of measurements. However, SOCA2024 appears to slightly underestimate b_{bp} at the deep b_{bp} maximum, likely due to the relatively coarse vertical resolution of the SOCA2024 outputs, which may miss the peak at certain depths in the shallow layer from which the entire vertical profile is derived. In the Southern Ocean, SOCA2024 b_{bp} values are consistent with the BGC-Argo float measurements, capturing the seasonal increase in b_{bp} from December to April in the 0–100 m layer. Similarly, in the North Atlantic Subpolar Gyre, the retrieved b_{bp} values align well with float measurements over the three-year time series. SOCA2024 effectively captures the elevated b_{bp} values at depth during the months following the typical productive bloom period in spring in the North Atlantic. For the Atlantic sub-Equatorial waters, SOCA2024 accurately retrieves the characteristic increase in b_{bp}

at relatively shallow depths (~50m). In both the Western and Eastern Mediterranean Sea, SOCA2024 estimations and float measurements show very strong consistency, reflecting the same seasonal variations and coherent vertical distributions throughout the year, including deep mixed conditions, bloom periods, and ultra-oligotrophic conditions.

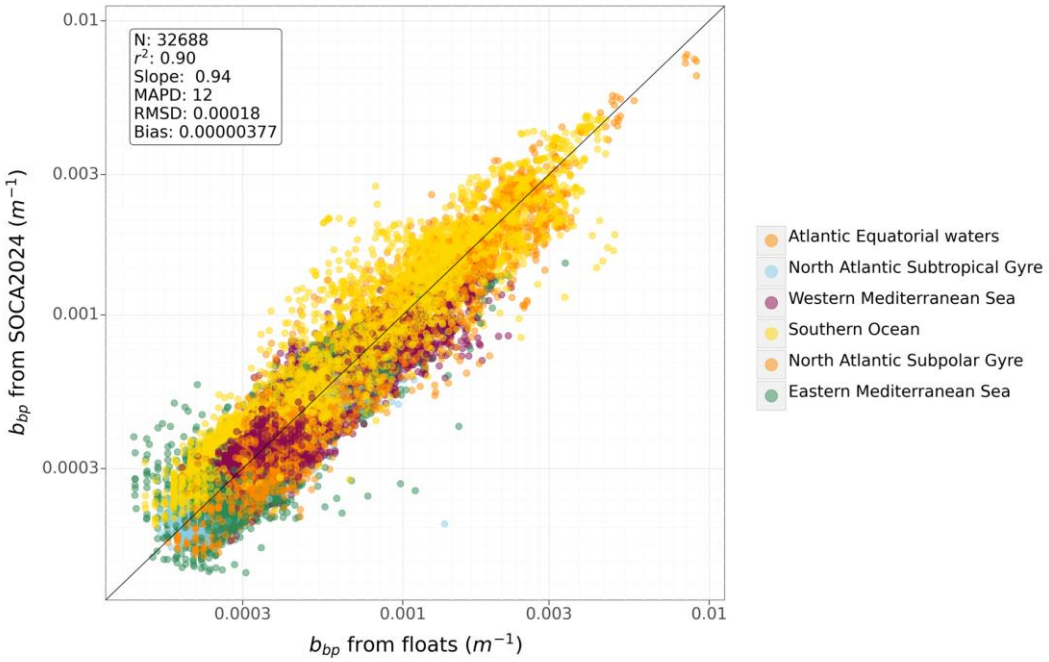


Figure 12: Comparison of the b_{bp} values retrieved by SOCA2024 to the reference b_{bp} measurements acquired by six BGC-Argo floats not integrated in the training and validation databases. The colour code indicates the oceanic basin in which the BGC-Argo floats were deployed (see geographic position in Figure 3). The 1:1 line is shown as black and the statistics details are provided.

Table 8: Statistics of comparison (RMSD: Root Mean Squared Difference, MAPD: Median Absolute Percent Difference, r^2: determination coefficient of the linear regression between estimates and observations) of the b_{bp} values predicted by SOCA2016, SOCA2020 and SOCA2024

Validation Dataset	Metrics	SOCA2016	SOCA2020	SOCA2024
Validation dataset (20% of the initial entire database)	r^2	0.78	0.90	**0.90**
	slope	0.8	0.89	**0.90**
	MAPD	21	11	**11**
	RMSD		0.00026	**0.00025**
Complete independent dataset (three BGC-Argo floats time series for SOCA2016 and SOCA2020 and six BGC-Argo time series for SOCA2024)	r^2	0.78	0.91	**0.9**
	slope	0.8	0.95	**0.94**
	MAPD	22	12.5	**12**
	RMSD		0.00026	**0.00018**
Independent database (North Atlantic Subtropical Gyre – WMO=6901472)	r^2	0.81	0.93	**0.94**
	slope	0.81	0.94	**0.92**
	MAPD	12	8	**8**
	RMSD		0.00007	**0.00006**
Independent database (Austral Ocean – WMO=6901493)	r^2	0.72	0.92	**0.93**
	slope	0.84	0.89	**0.89**
	MAPD	23	20	**17**
	RMSD		0.00035	**0.00030**
Independent database (North Atlantic Subpolar Gyre – WMO=6901523)	r^2	0.85	0.93	**0.93**
	slope	0.85	0.89	**1.02**
	MAPD	21	16	**15**
	RMSD		0.00031	**0.00027**
Independent database (Atlantic Equatorial Waters – WMO=4903459)	r^2			**0.86**
	slope			**0.91**
	MAPD			**11**
	RMSD			**0.00013**
Independent database (Western Mediterranean Sea – WMO=6901491)	r^2			**0.83**
	slope			**0.82**
	MAPD			**12**
	RMSD			**0.00015**
Independent database (Eastern Mediterranean Sea – WMO=6901773)	r^2			**0.80**
	slope			**0.93**
	MAPD			**12**
	RMSD			**0.00009**

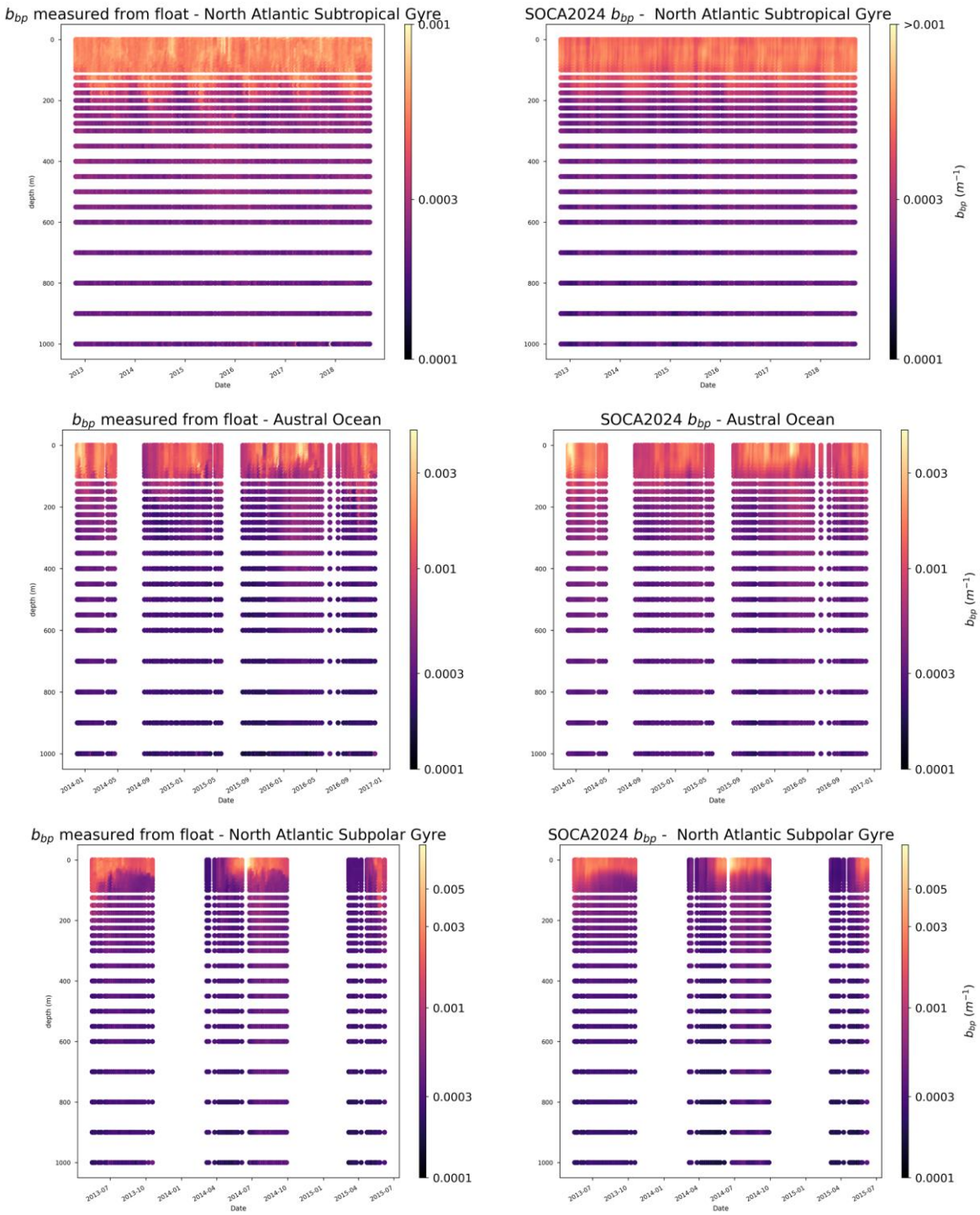


Figure 13: Comparison of the reference b_{bp} measurements acquired by BGC-Argo floats (left) with the values predicted by SOCA2024 (right) for the BGC-Argo floats deployed in the North Atlantic Subtropical Gyre, the Southern Ocean and the North Atlantic Subpolar Gyre.

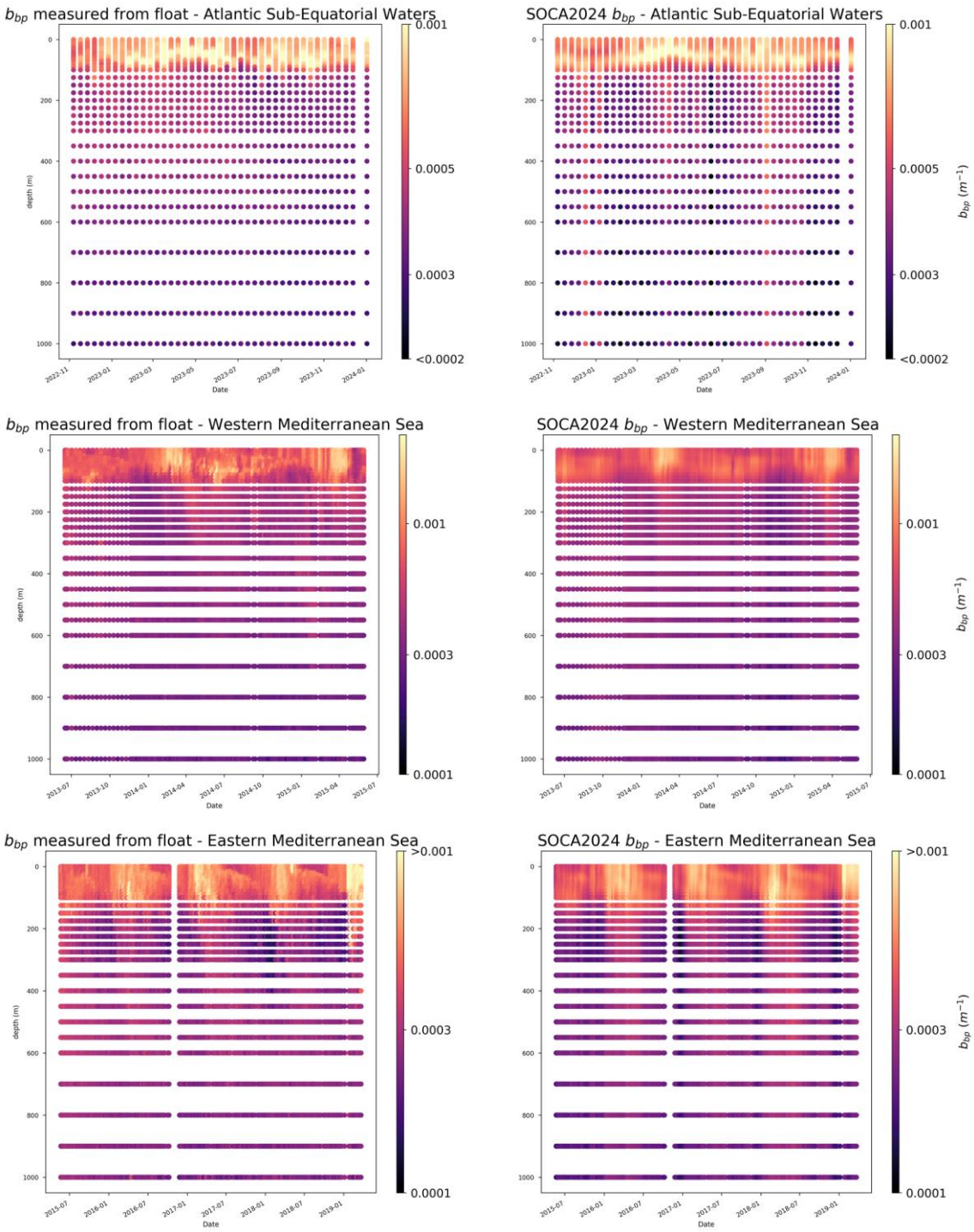


Figure 14: Comparison of the reference b_{bp} measurements acquired by BGC-Argo floats (left) with the values predicted by SOCA2024 (right) for the BGC-Argo floats deployed in the Atlantic Equatorial Waters, and the Western and Eastern Mediterranean Sea.

IV.2 Chlorophyll-a concentration

The accuracy of the derived Chl product is evaluated here against multiple high-quality *in situ* measurements to ensure robustness. Validation is performed against both the BGC-Argo database (20% subset and 6 independent floats) and highly accurate HPLC measurements.

IV.2.1 Validation against a 20% random subset of the BGC-Argo database

The performance of SOCA2024 is evaluated using 20% of the dataset chosen randomly from the BGC-Argo Chl database. The comparison between SOCA2024 Chl retrieval against the BGC-Argo Chl as reference measurements is presented in Figure 15. The global error is ~29% (i.e., median absolute percent difference). SOCA2024 method predicts Chl without much bias according to the depth of retrieval. A slight broadening of scatter can be visible where Chl are less than 0.01 mg m^{-3}, which is below the sensor detection limit of BGC-Argo float. Figure 15 shows that most of SOCA2024 Chl values were retrieved with good accuracy.

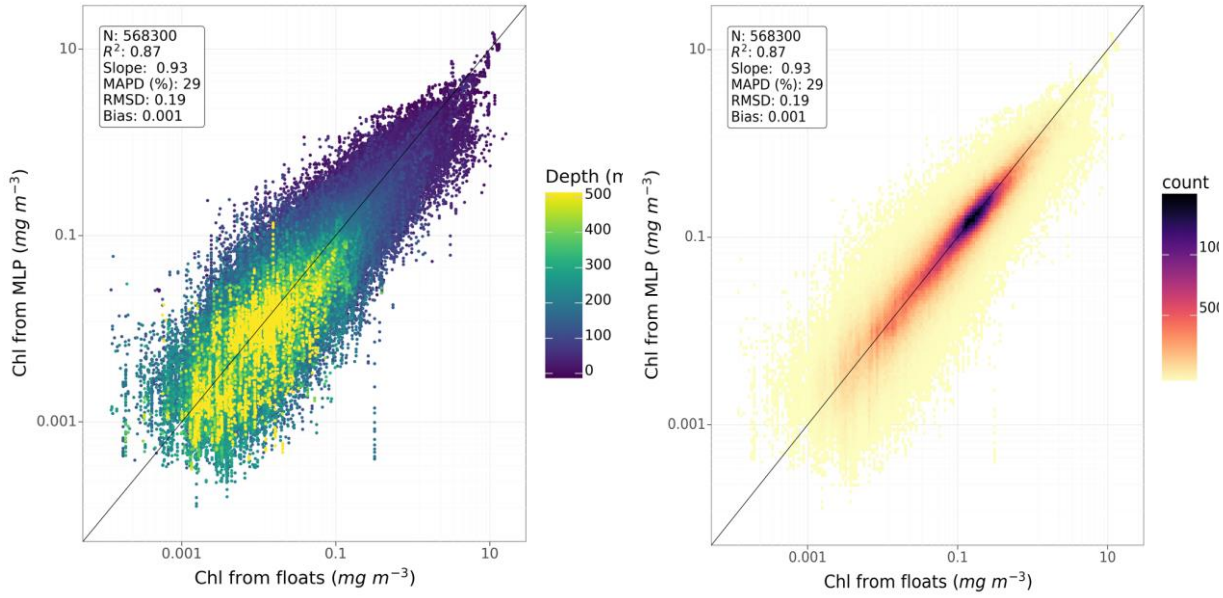


Figure 15: Comparisons of Chl values retrieved by SOCA2024 to the reference measurements acquired by BGC-Argo profiling floats from validation dataset (11,340 profiles). The left panel shows the colours in corresponding depths levels and the right panel shows the density scatter with colours as the number of points in each bin. The 1:1 line is shown as black and statistics details are also provided. Note that we removed data from floats with Chl less than 0.0001 mg m^{-3}.

The spatial distribution of the accuracy (median percent difference of Chl between SOCA2024 and BGC-Argo) is shown in Figure 16. The figure shows that Chl was retrieved with slightly less accuracy in high latitude environments, which are areas where more seasonal and spatial variability in the vertical distribution of Chl was observed. However, it is important to note that there is no systematic geographical bias evident in Figure 16.

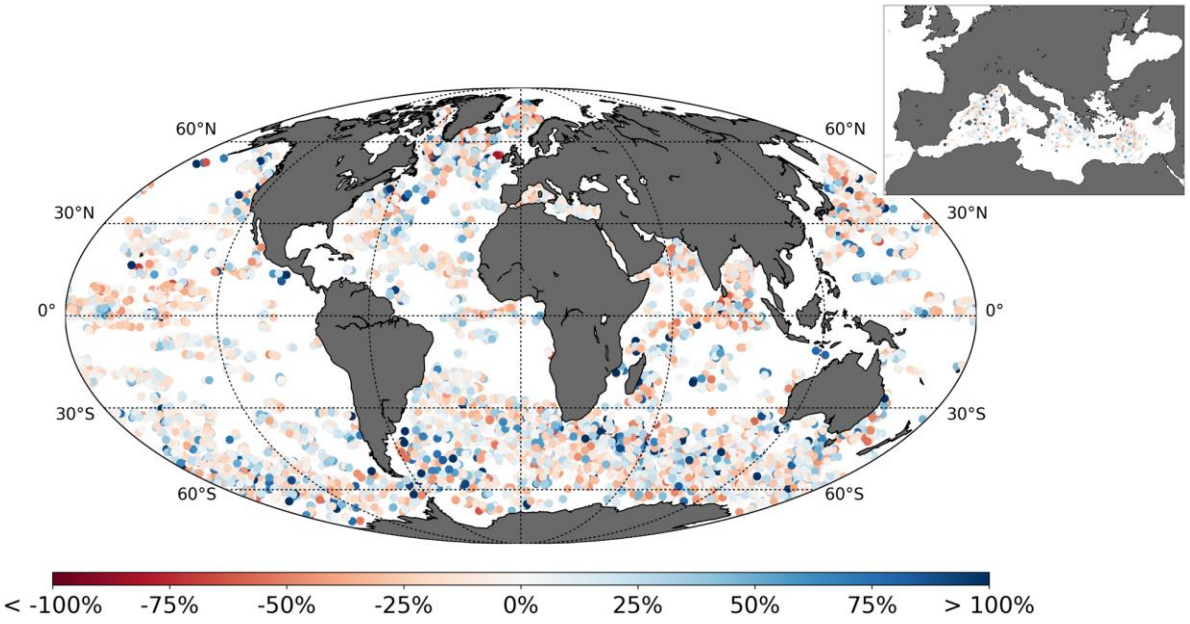


Figure 16: Geographical distributions of median percent difference for each profile between SOCA2024 Chl estimates and BGC-Argo Chl data. Zoom of the Mediterranean Sea in the upper right panel.

IV.2.2 Validation against six independent BGC-Argo floats

To further validate and assess the accuracy of Chl retrieval from SOCA2024, six independent BGC-Argo profiling float time series were excluded from the database used for training and validation of the SOCA2024 method. These floats, originating from six different ocean basins with distinct hydrographic conditions, were set aside to form a completely independent dataset. Testing the SOCA2024 method on this independent dataset revealed similar accuracy to that observed with the original validation dataset, which comprised 20% of the initial database (Figure 17 and Table 9).

The multiyear time series of Chl profiles from BGC-Argo floats and predicted SOCA2024 Chl profiles for the six independent floats are shown in Figure 18 and Figure 19. The vertical patterns of Chl predicted by SOCA2024 are highly consistent with those observed by profiling floats. The comparison of the BGC Argo Chl and SOCA2024 Chl profiles, reveals that SOCA2024 is able to adequately reproduce the seasonal pattern of deepening and shallowing of Deep Chlorophyll Maximum (DCM). In the North Atlantic Subtropical gyre, multiyear data (WMO=6901472) provide a robust basis for validation. Comparison with the reference BGC-Argo profile shows that the SOCA2024 retrieval is very consistent and reproduces the annual and seasonal cycles, including the deepening of chlorophyll-a maximum. In the North Atlantic Equatorial waters (WMO = 4903459), the permanent and shallower DCM is consistently retrieved by SOCA2024 with respect to reference data in all seasons. From the Mediterranean Sea, we have selected two independent floats, one from the Eastern basin (WMO=6901773) and the other from the Western basin (WMO=6901491). The SOCA2024 model retrieval from these two basins is good and consistent with reference BGC-Argo for all the seasons. In the Mediterranean Sea, SOCA2024 Chl retrieval is underestimated when compared with the reference Chl measured by BGC-Argo floats. The prediction of SOCA2024 Chl for the two independent floats operating in high latitudes environments, one from the south (WMO=6901493) and the other from the north (WMO=6901523), are significantly consistent. Gaps or missing data are due to clouds or winter season.

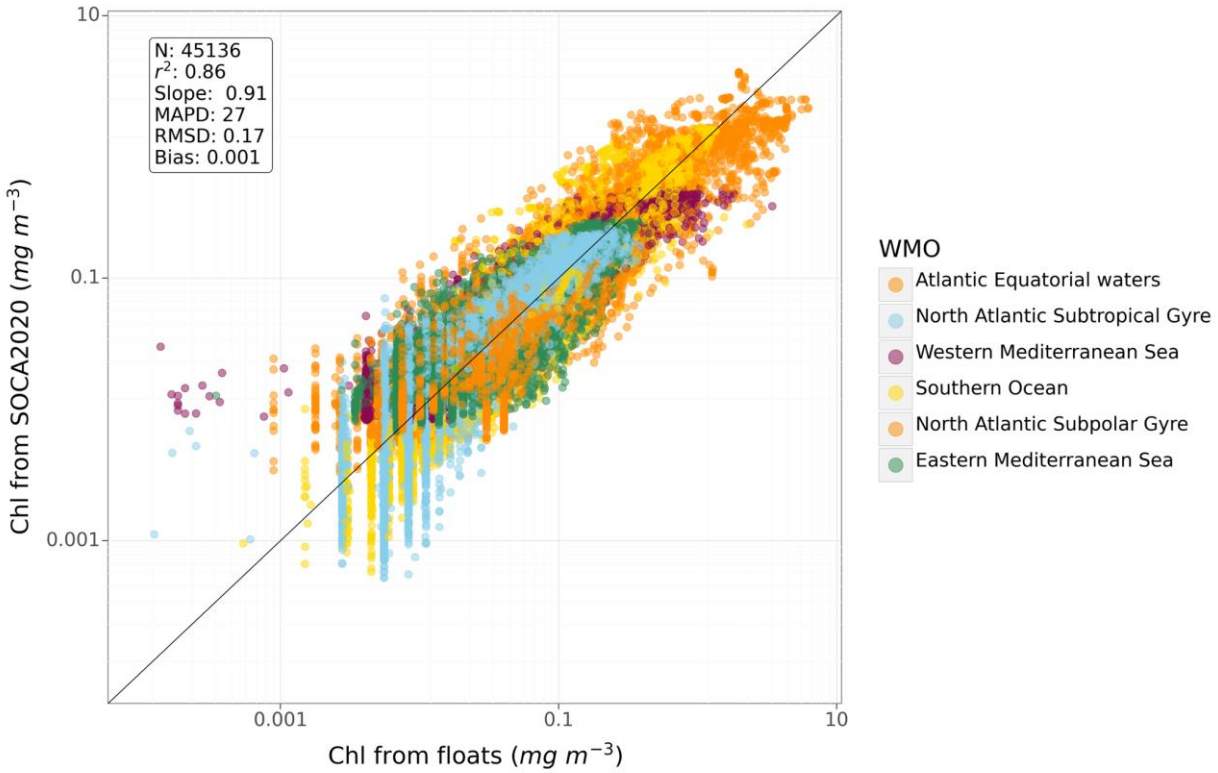


Figure 17: Comparison of Chl values retrieved by SOCA2024 to the reference Chl measurements acquired by six independent BGC-Argo floats (these floats are not integrated in training and validation datasets). The 1:1 line is shown as black dotted line and statistics details are provided.

Table 9: Statistics of comparison (RMSD: Root Mean Squared Difference, MAPD: Median Absolute Percent Difference, r^2: determination coefficient of the linear regression between estimates and observations) of Chl values predicted by SOCA2021 and SOCA2024 using BGC-Argo Chl as reference data.

Validation Dataset	Metrics	SOCA2021	SOCA2024
Validation dataset (20% of the initial entire database)	r^2	0.82	0.83
	slope	0.92	0.82
	MAPD	29	30
	RMSD	0.31	0.19
Complete independent dataset (the six BGC-Argo floats time series)	r^2	0.80	0.85
	slope	0.92	0.89
	MAPD	30	27
	RMSD	0.28	0.17
Independent database (North Atlantic Subtropical Gyre – WMO=6901472)	r^2	0.80	0.86
	slope	0.70	1.03
	MAPD	25	25
	RMSD	0.04	0.03
Independent database (Austral Ocean – WMO=6901493)	r^2	0.73	0.88
	slope	0.81	1.02
	MAPD	33	29
	RMSD	0.33	0.12
Independent database (North Atlantic Subpolar Gyre – WMO=6901523)	r^2	0.78	0.86
	slope	0.94	0.90
	MAPD	42	35
	RMSD	0.85	0.50
Independent database (Atlantic Equatorial Waters – WMO=4903459)	r^2		0.86
	slope		0.77
	MAPD		35
	RMSD		0.09
Independent database (Western Mediterranean Sea – WMO=6901491)	r^2		0.91
	slope		0.80
	MAPD		26
	RMSD		0.09
Independent database (Eastern Mediterranean Sea – WMO=6901773)	r^2		0.82
	slope		0.79
	MAPD		23
	RMSD		0.03

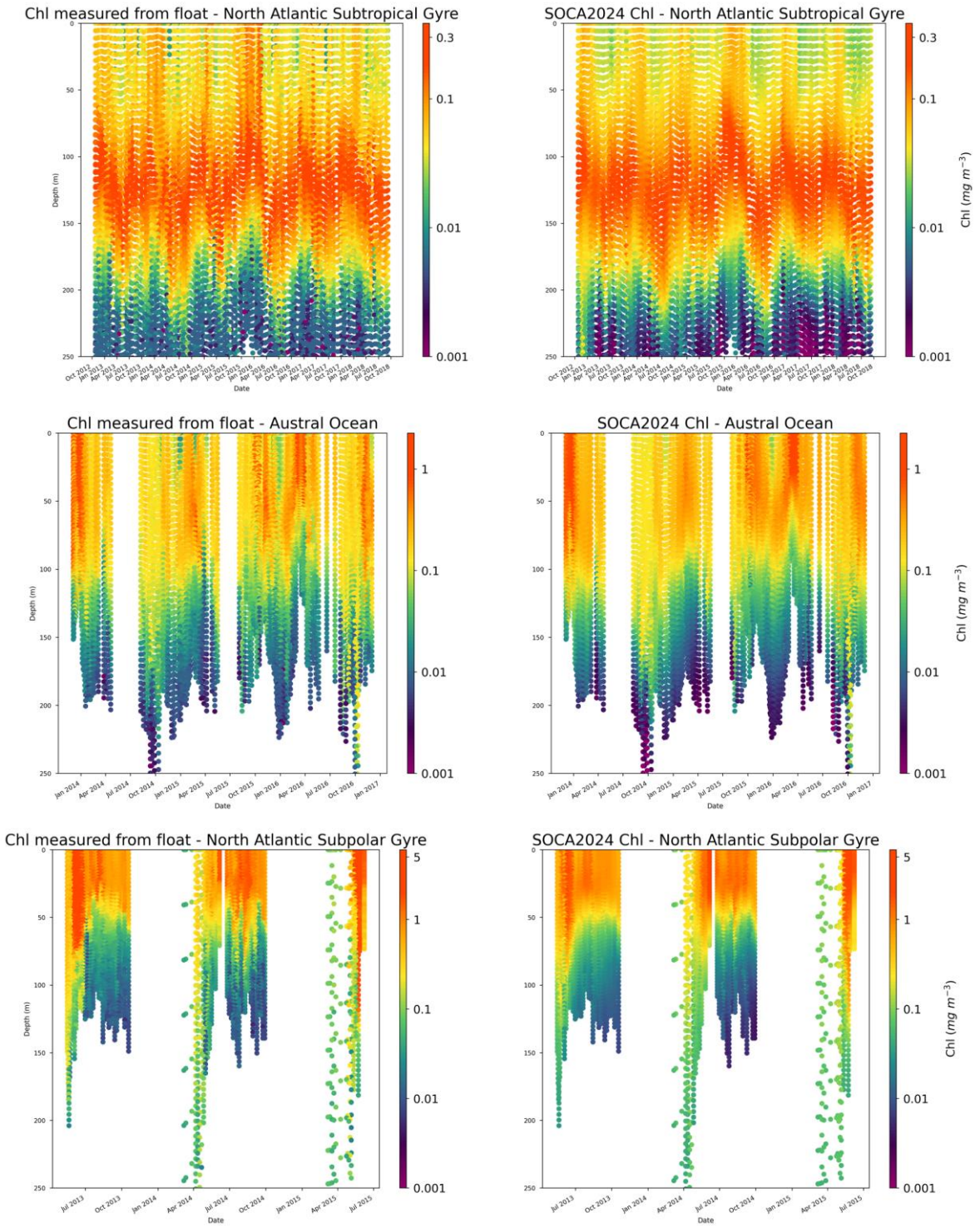


Figure 18: Comparison of the reference Chl measurements acquired by BGC-Argo floats (left) with the values predicted by SOCA2024 (right) for the BGC-Argo floats deployed in the North Atlantic Subtropical Gyre, the Southern Ocean and the North Atlantic Subpolar Gyre.

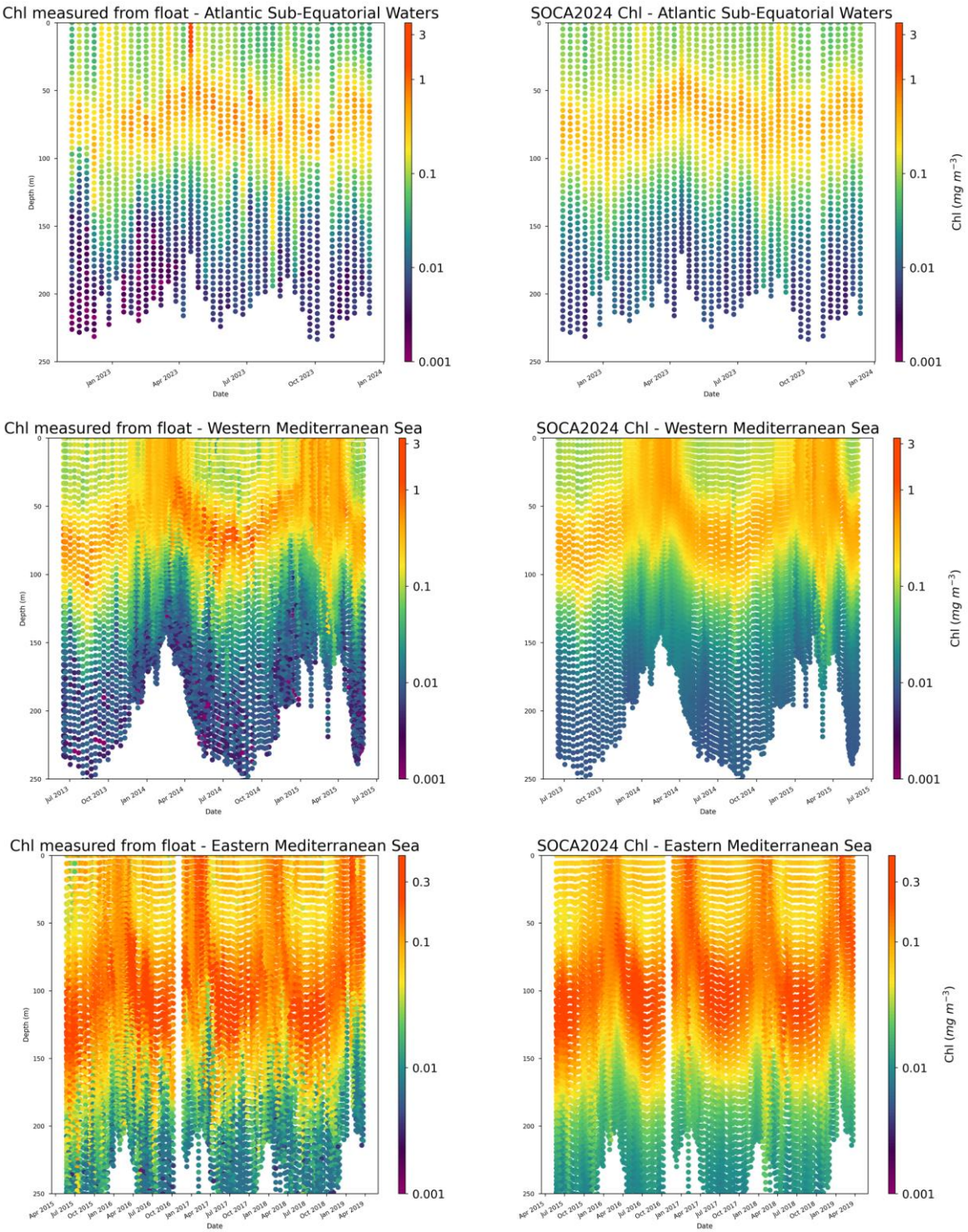


Figure 19: Comparison of the reference Chl measurements acquired by BGC-Argo floats (left) with the values predicted by SOCA2024 (right) for the BGC-Argo floats deployed in the Atlantic Equatorial Waters, and the Western and Eastern Mediterranean Sea.

IV.2.3 Validation against HPLC database

The HPLC pigment database from SAPIGH, the HPLC analytical service hosted at the Institut de la Mer de Villefranche (IMEV) from the Laboratoire d'Océanographie de Villefranche (LOV), and the MAREDAT database (Peloquin et al., 2013) were gathered to create a complete HPLC database to validate the SOCA2024 Chl and to assess the accuracy of the inference of the vertical distribution of Chl in the **MULTIOBS_GLO_BIO_BGC_3D_REP_015_010** product. The HPLC dataset is composed of 2,674 Chl profiles with a bathymetry deeper than 1000m collected since 1998 (see orange dots in Figure 20). From this database, 1,818 Chl profiles have a concomitant matchup of satellite R_{rs}, PAR, SLA, ADT and hydrological data from weekly products. These profiles are used to evaluate the SOCA2024 Chl vertical distribution to the reference HPLC Chl profile.

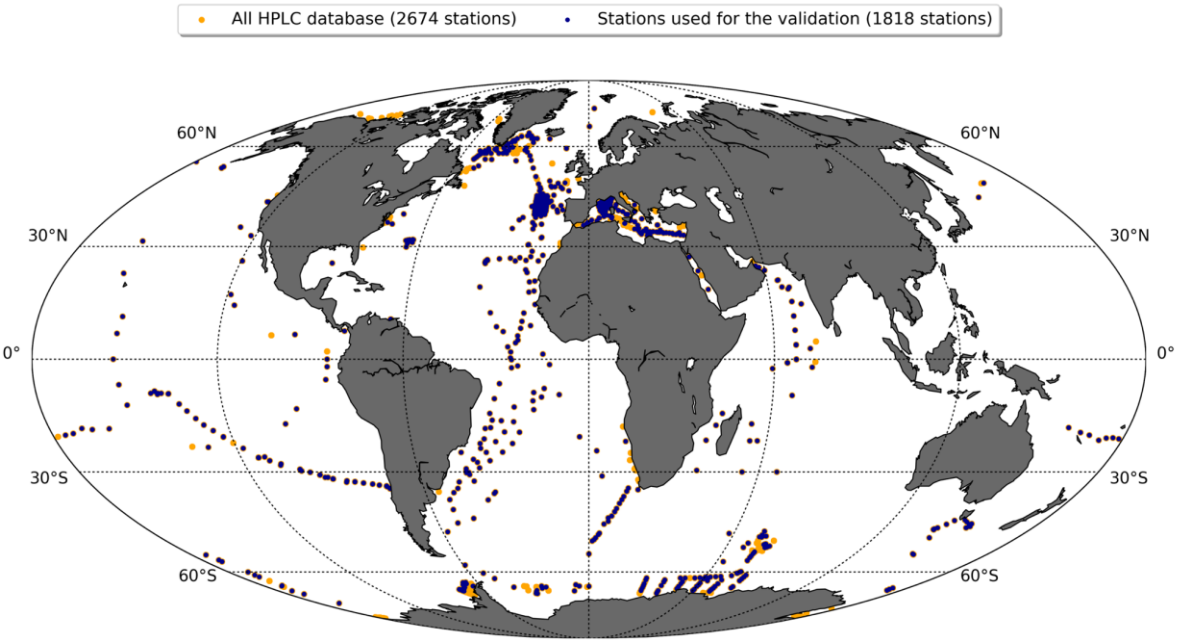


Figure 20: Geographical distribution of the 2,674 available HPLC profiles (orange dots). Of these, 1,818 profiles have corresponding ocean colour, SLA, and hydrological weekly data (blue dots). These 1,818 profiles have been used to evaluate the MULTIOBS_GLO_BIO_BGC_3D_REP_015_010 Chl product.

The comparison of SOCA2024 Chl and HPLC Chl as reference data has been performed (Figure 21, left panel). The validation against HPLC Chl data shows a slope of 0.70, with a significant clustering of points around the 1:1 line and a mean absolute percent difference (MAPD) of 40%. The large scatter is primarily attributed to the inherent uncertainties in retrieving Chl from chlorophyll-a fluorescence, the measurement method used by BGC-Argo profiling floats, and comparing it with 'true' Chl derived from HPLC. Chl estimation from fluorescence measurements leads to seasonal and regional uncertainties associated with changes in phytoplankton community composition and their physiological state (Roesler et al., 2017). High latitudes environments, particularly the Southern Ocean, are known for overestimations in Chl due to the fluorescence-based estimations. However, by using the upgraded BGC-Argo Chl dataset in D-mode, which includes corrections to address this bias, the overestimation has been reduced in SOCA2024, as can be seen in Figure 22 (right panel), which presents the validation against

HPLC in the Southern Ocean (latitude < 60°S) for both the previous model (SOCA2021, Uitz et al., 2006) and the new SOCA2024 model.

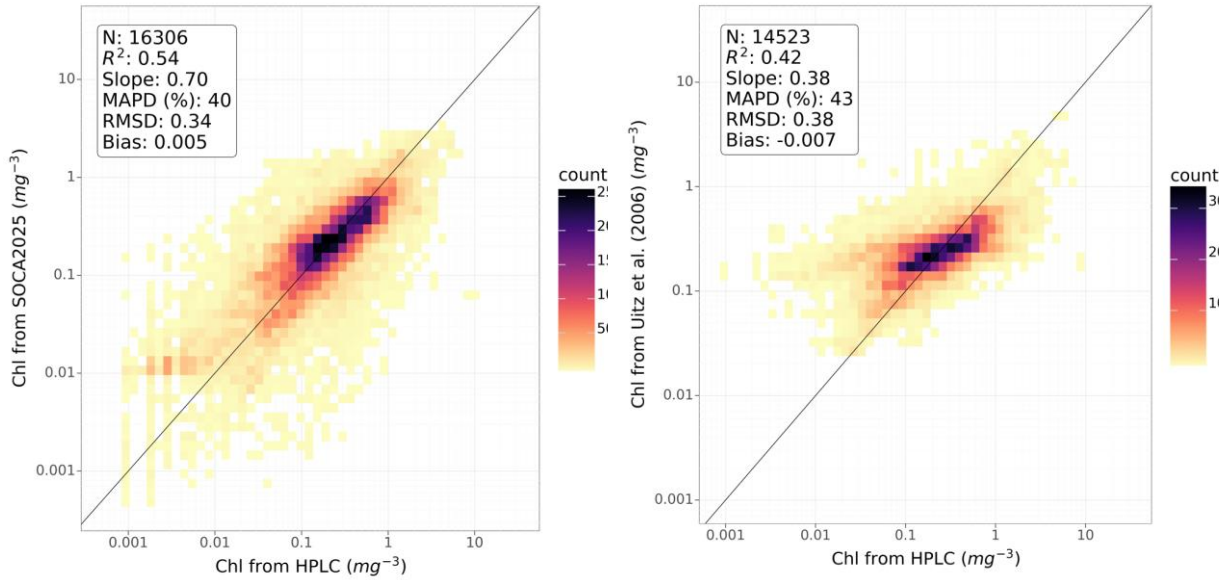


Figure 21: Comparison of Chl retrieved by SOCA2024 with reference HPLC Chl from MAREDAT and LOV database. Data are colour coded according to the amount of data in each bin. The SOCA2024 model evaluation is presented in the left panel, while the Uitz et al. (2006) parameterization is in the right panel. Statistics (computed using data with HPLC Chl>0.01 mg m^{-3}) are indicated for evaluation.

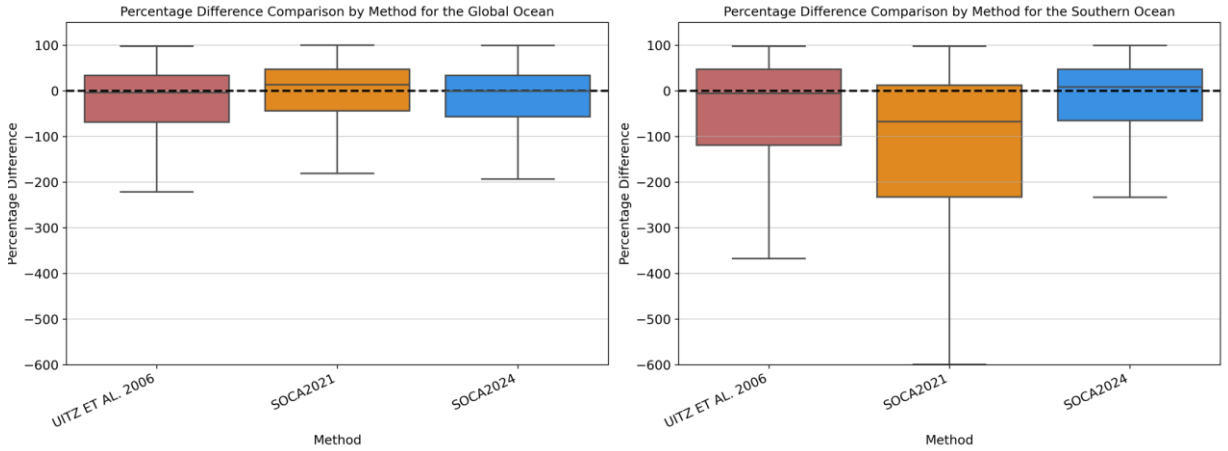


Figure 22: Comparison of the accuracy of Chl retrieved by Uitz et al. (2006), SOCA2021, and SOCA2024 methods evaluated from the difference (in percentage) with reference HPLC Chl for the Global Ocean (left panel) and the Southern Ocean (latitude < 60°S; right panel).

More generally, a comparison of SOCA2024 versus the model of Uitz et al. (2006) used for the first release of the product has been performed against the reference HPLC database (Figure 21 and Figure 22). The data which were used for the Uitz et al. (2006) parameterization method have been removed from this analysis. The scatterplot of SOCA2024 Chl versus HPLC Chl shows better clustering of data

points near the 1:1 line and better statistics while comparing with the scatterplot of Chl from Uitz et al. (2006) versus HPLC Chl (Figure 21).

The distribution of the median percent difference of each profile when compared to the HPLC reference dataset reveals several spatial patterns (Figure 23). Although the overestimation of Chl in the Southern Ocean has been reduced compared to the third release of the product (using SOCA2021; see right panel on Figure 22) due to the upgraded BGC-Argo Chl dataset, an overestimation remains noticeable in the North Atlantic (Figure 23).

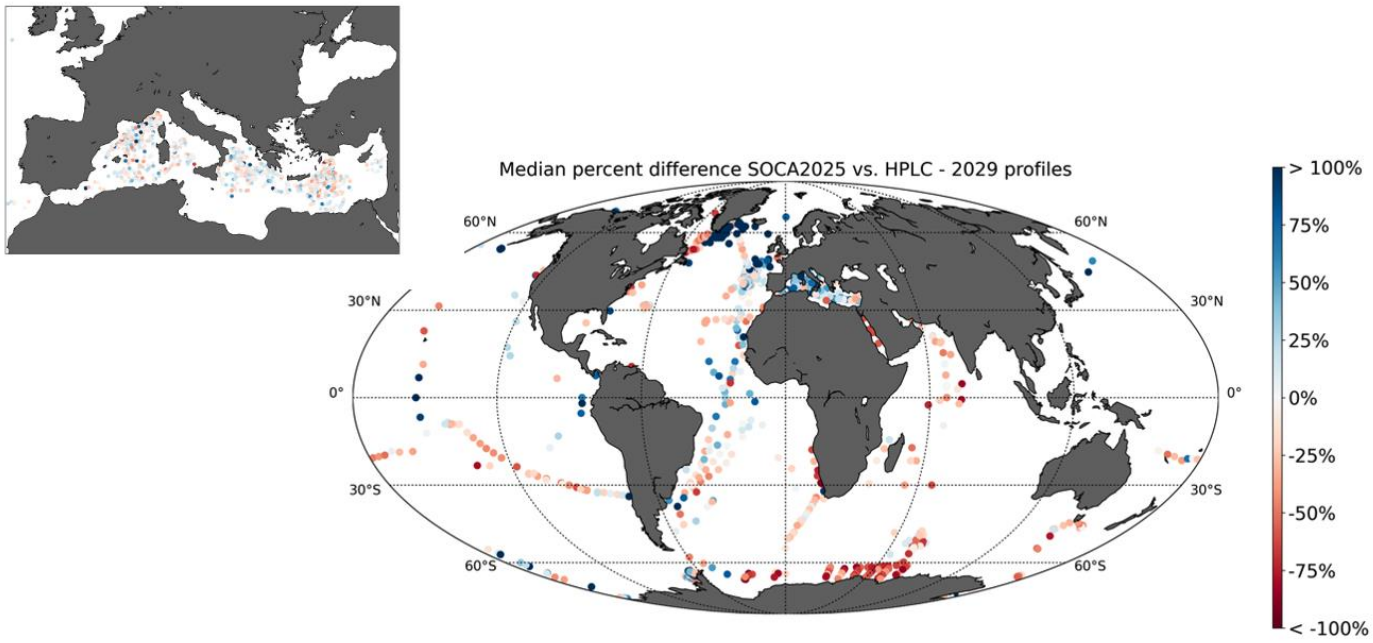


Figure 23: Geographical distribution of the median percent difference of each profile between SOCA2024 estimated Chl and HPLC reference Chl data. Zoom of the Mediterranean Sea in the upper right panel.

IV.3 Light variables (PAR, ED380, ED412 and ED490)

A comprehensive validation framework was implemented to evaluate the accuracy of the four light models. Firstly, the model outputs were compared with the primary 20% validation dataset (Section IV.3.1). This was followed by validation against data from four independent BGC-Argo floats, each located in a different ocean basin (Section IV.3.2), and subsequently against the independent SeaBASS database (Section IV.3.3). Lastly, additional proxies derived from SOCA-light-2025 products were employed to further assess the model's predictive performance (Section IV.3.4). The SOCA-light-2025 models represent a substantial leap forward in performance. Detailed analysis of the accuracy metrics confirms that these new models are significantly better than all previous iterations of the SOCA-light series. This breakthrough indicates enhanced reliability and precision in the model's predictions or classifications. A direct comparison illustrating the magnitude of these improvements is presented in Table 10, which highlights the superior results achieved across all relevant evaluation measures.

IV.3.1 Validation against a 20% random subset of the BGC-Argo database

The SOCA-light-2025 models were validated using 20% of the dataset randomly extracted from the BGC-Argo database, originating from a large diversity of oceanic regions. The comparison between modelled SOCA-light-2025 variables and BGC-Argo measurements (PAR, ED380, ED412, and ED490) is presented in Figure 24. Overall, there is a very good agreement between the predicted and the measured light variables. The density scatterplot reveals a close clustering of points along the identity line over more than five orders of magnitude. Statistical metrics extracted from linear regression between the modelled and observed PAR values reveal slope, r^2, RMSD and MAPD values of 0.98, 0.96, 80.03 µmol quanta m^{-2} s^{-1}, and 33.68%, respectively (Figure 24A). Validation for all three ED models shows satisfactory performances. The modelled ED380 profiles exhibit RMSD and MAPD values of 0.04 W m^{-2} nm^{-1}, and 37.31%, respectively, when compared with their measured counterparts (Figure 24B). Similarly, the MAPD values for ED412 and ED490 were 37.24% and 33.32%, respectively (Figure 24 and Table 10).

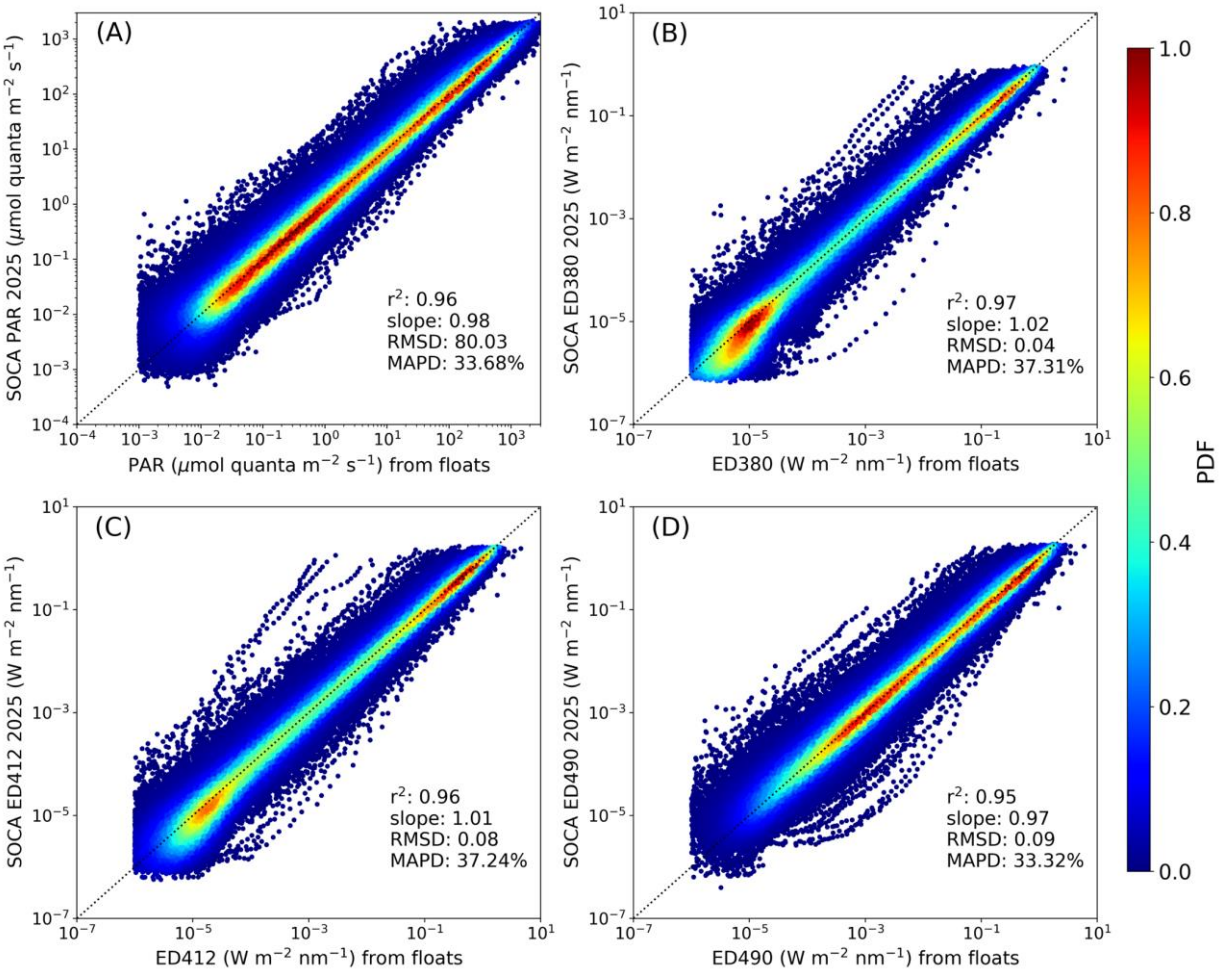


Figure 24: Scatterplots between light variables (PAR, ED380, ED412, and ED490) modelled by the SOCA-light-2025 models versus their corresponding BGC-Argo measurements: PAR (A); ED380 (B); ED412 (C); ED490 (D). This validation was performed using 20% of profiles randomly selected from the total database. The colour code scales the probability density function (PDF). The identity line is represented by the 1:1 black dotted line.

IV.3.2 Validation against four independent BGC-Argo floats

An independent validation was performed using four BGC-Argo floats from distinct oceanic regions, namely the North Atlantic Subtropical Gyre (NASTG), the Eastern Mediterranean Sea (EMS), the Southern Ocean (SO), and the North Atlantic Subpolar Gyre (NASPG). The profiles for each region originated from a single float with a unique WMO, none of which were included in the training and validation databases. Validation results for each oceanic region are presented in Figure 25 and Table 10. The scatterplot of PAR derived by the model shows strong agreement with PAR measured by the BGC-Argo floats for all four regions (Figure 25A). Statistical error estimators computed between modelled and observed PAR profiles for all four regions together show slope, r^2, RMSD, and MAPD values of 1.02, 0.96, 73.41 µmol quanta m^{-2} s^{-1}, and 27.34%, respectively. These statistical error estimators of PAR are comparable with the statistics obtained on 20% of the validation database (Figure 24A). For ED380 (Figure 25B), the slope, r^2, RMSD, and MAPD values are 1.04, 0.97, 0.04 W m^{-2} nm^{-1}, and 39.13%, respectively. For ED412 (Figure 25C), the same statistical metrics yield values of 1.03, 0.97, 0.06 W m^{-2} nm^{-1}, and 34.47%, respectively. Finally, for ED490 (Figure 25D), the metrics take values of 1.01, 0.96, 0.08 W m^{-2} nm^{-1}, and 29.04%, respectively.

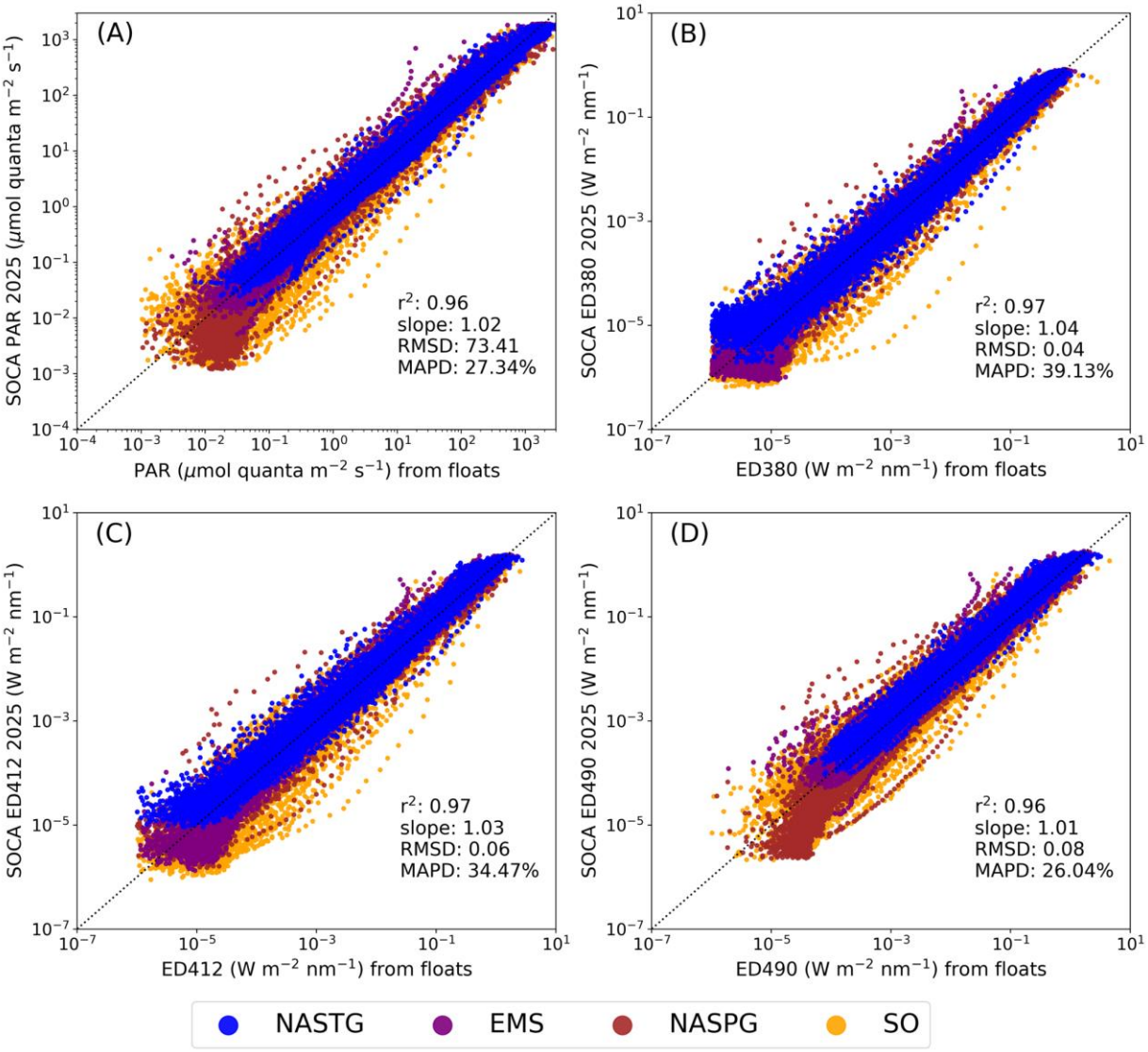


Figure 25: Scatterplots illustrating the comparison between SOCA-light-2025 modelled variables (PAR, ED380, ED412, and ED490) and their corresponding BGC-Argo measurements collected by the four independent floats. The subplots display: PAR (A), ED380 (B), ED412 (C), ED490 (D). Each colour represents a specific float: blue for NASTG, purple for EMS, brown for NASPG, orange for SO. The identity line is represented by the 1:1 black dotted line.

Table 10: Statistics (RMSD: Root Mean Squared Difference, MAPD: Median Absolute Percent Difference, r^2: determination coefficient of the linear regression between estimates and observations) comparing of light variable predictions from the SOCA-light and SOCA-light-2025 models (specifically PAR, ED380, ED412 and ED490) to observed reference data from BGC-Argo and SeaBASS.

Validation Dataset	Metrics	SOCA-light				SOCA-light 2025			
		PAR	ED380	ED412	ED490	PAR	ED380	ED412	ED490
Validation dataset (20% of the initial entire database)	r^2	0.96	0.97	0.96	0.95	**0.96**	**0.97**	**0.96**	**0.95**
	slope	1.01	1.01	1.00	0.97	**0.98**	**1.02**	**1.01**	**0.97**
	RMSD	76.42	0.04	0.08	0.09	**80.03**	**0.04**	**0.08**	**0.09**
	MAPD	37.41	39.01	39.47	37.05	**33.68**	**37.31**	**37.24**	**33.32**
Complete independent dataset (the four BGC-Argo floats time series)	r^2	0.96	0.97	0.97	0.95	**0.96**	**0.97**	**0.97**	**0.96**
	slope	1.03	1.04	1.03	1.03	**1.02**	**1.04**	**1.03**	**1.01**
	RMSD	72.86	0.03	0.07	0.09	**73.41**	**0.04**	**0.06**	**0.08**
	MAPD	30.50	40.99	36.67	29.86	**27.34**	**39.13**	**34.47**	**26.04**
Independent database (North Atlantic Subtropical Gyre – WMO=6901472)	r^2	0.98	0.98	0.98	0.98	**0.98**	**0.98**	**0.98**	**0.98**
	slope	0.99	0.96	0.95	0.98	**0.99**	**0.94**	**0.94**	**0.98**
	RMSD	73.09	0.04	0.08	0.09	**82.89**	**0.04**	**0.08**	**0.10**
	MAPD	21.50	28.72	26.66	21.53	**22.65**	**29.12**	**27.35**	**20.01**
Independent database (Southern Ocean – WMO=6901493)	r^2	0.91	0.95	0.93	0.91	**0.94**	**0.96**	**0.95**	**0.94**
	slope	0.99	1.03	1.03	0.99	**0.99**	**1.06**	**1.04**	**0.98**
	RMSD	88.63	0.04	0.08	0.10	**73.35**	**0.05**	**0.07**	**0.09**
	MAPD	54.37	51.52	54.33	51.47	**39.52**	**48.36**	**45.73**	**38.56**

Table 10: (continue)

Validation Dataset	Metrics	SOCA-light				SOCA-light 2025			
		PAR	ED380	ED412	ED490	PAR	ED380	ED412	ED490
Independent database (Eastern Mediterranean Sea – WMO=6901773)	r^2	0.98	0.98	0.98	0.98	**0.98**	**0.98**	**0.98**	**0.98**
	slope	1.02	1.08	1.06	1.01	**1.01**	**1.08**	**1.04**	**0.99**
	RMSD	63.20	0.03	0.06	0.08	**63.13**	**0.03**	**0.06**	**0.07**
	MAPD	21.42	39.89	29.04	22.78	**18.04**	**35.15**	**24.83**	**17.87**
Independent database (North Atlantic Subpolar Gyre – WMO=6901523)	r^2	0.93	0.96	0.95	0.92	**0.93**	**0.95**	**0.94**	**0.92**
	slope	1.08	1.02	1.02	1.09	**1.07**	**1.03**	**1.04**	**1.09**
	RMSD	64.83	0.02	0.04	0.06	**74.45**	**0.03**	**0.05**	**0.06**
	MAPD	58.79	50.82	51.75	57.63	**51.55**	**48.06**	**51.29**	**54.96**
Independent SeaBASS database	r^2	0.88	0.82	0.81	0.85	**0.86**	**0.88**	**0.86**	**0.86**
	slope	1.00	1.00	1.00	0.99	**1.02**	**0.98**	**0.99**	**0.97**
	RMSD	101.25	0.11	0.18	0.21	**85.48**	**0.10**	**0.16**	**0.19**
	MAPD	65.48	76.30	76.07	62.32	**54.39**	**63.42**	**62.29**	**48.95**

North Atlantic Subtropical Gyre

The NASTG is an oligotrophic environment characterized by low surface nutrients, low Chl, and the presence of a permanent deep chlorophyll maximum (DCM), generally found below 100 m (Bock et al., 2022; Cornec et al., 2021). The multi-year time series (more than six years of measurement) of the vertical distribution of light variables (PAR, ED380, ED412, and ED490) measured by the NASTG BGC-Argo float (WMO = 6901472) and modelled by SOCA-light-2025 are presented in Figure 26 for a direct comparison. Overall, the SOCA-light-2025 models clearly reproduce, in a smoother way, the seasonal and vertical trends revealed by the float measurements. The SOCA-light-2025 models capture even subtle changes in the general trends of light variables, as evidenced by the less pronounced light penetration observed and reproduced by the model at the end of 2015. As well as reproducing the seasonal trends satisfactorily, the magnitude of the signals is retrieved well by the models for the four variables.

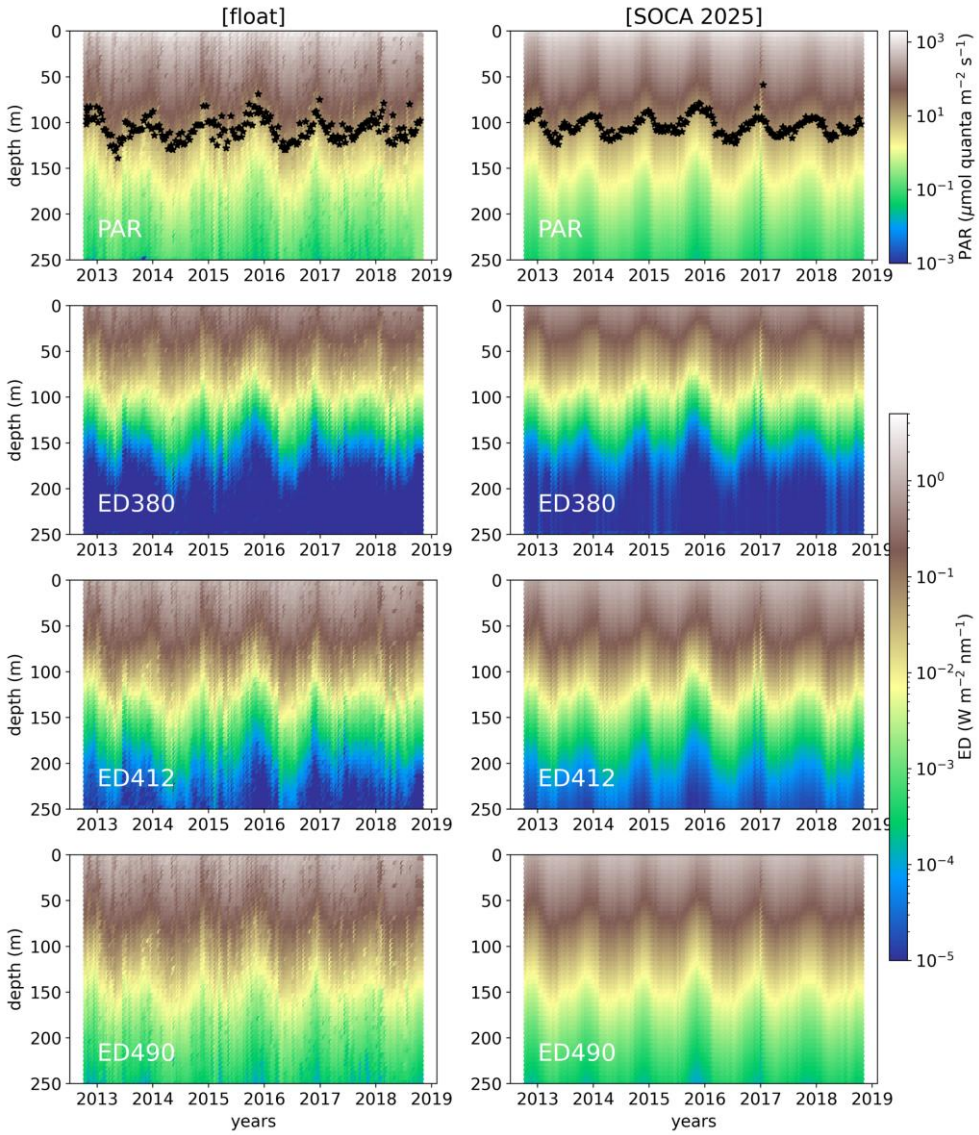


Figure 26: Time series of the vertical distribution of the four light variables in the North Atlantic Subtropical Gyre (NASTG), as measured by BGC-Argo float with WMO 6901472 (left column) and modelled by SOCA-light-2025 (right column). The variables in each subplot are indicated by text in the corresponding subplots. The black stars indicate the depth at which instantaneous PAR value = 15 µmol quanta m^{-2} s^{-1}.

Southern Ocean

Over four years, the BGC-Argo float (WMO = 6901493) traveling eastwards (from 5°E to 83°E) and between 40°S and 50°S in the SO underwent the typical bio-physical conditions prevailing in the area. Overall, it captured four phytoplankton blooms and was regularly trapped or influenced by mesoscale features or fronts. The multi-year time series of the vertical distribution of light variables (PAR, ED380, ED412, and ED490) measured by this float and the SOCA-light-2025 modelled light variables are presented and compared in Figure 27. The gaps in the time series during the southern-hemisphere winter months are due to the unavailability of ocean colour matchups resulting from cloud coverage during this period. In general, as for the NASTG, the SOCA-light-2025 models reproduce the seasonal

and vertical trends of the float measurements in a smoother way. In addition to reproducing the seasonal trends, the magnitude of the retrieved light variables is of the same order of magnitude as the measurements for the four light variables.

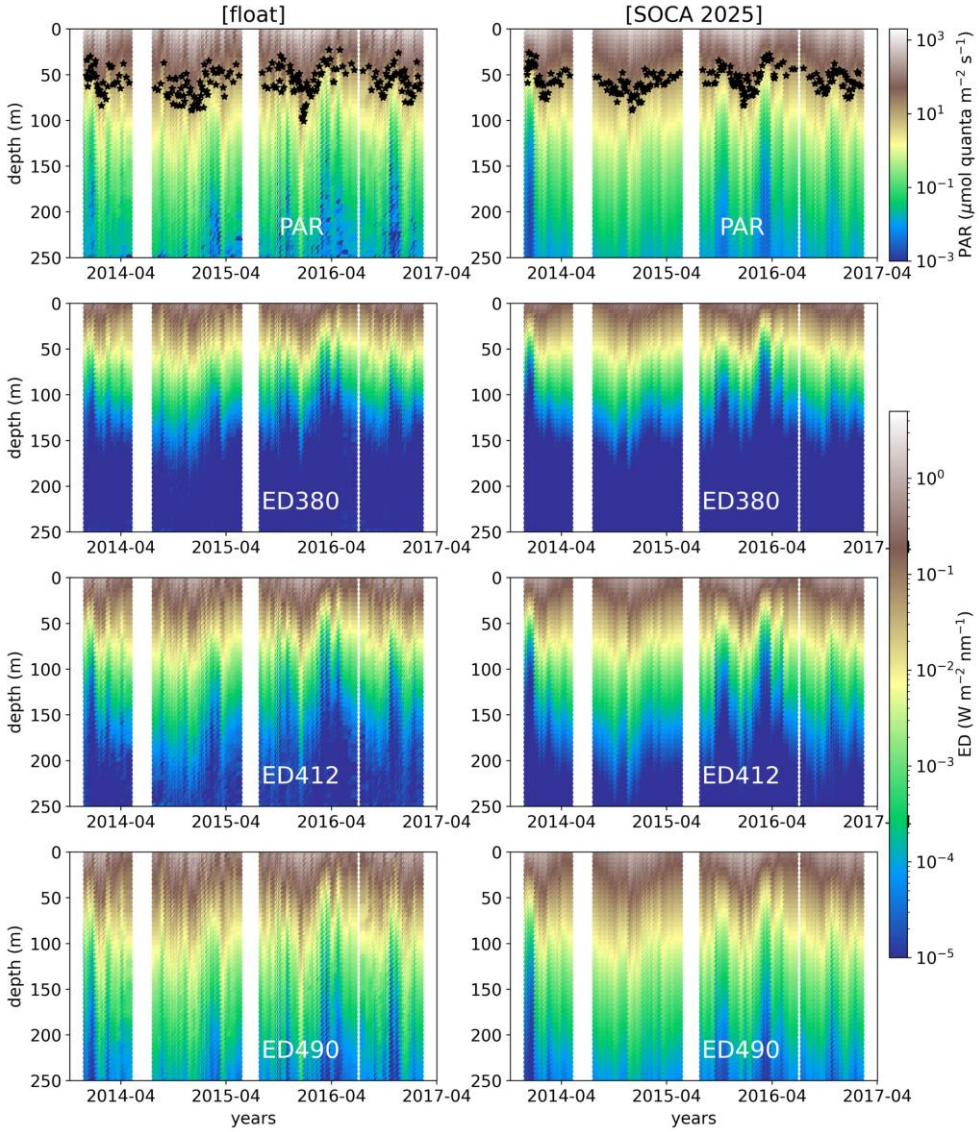


Figure 27: Time series of the vertical distribution of the four light variables in the Southern Ocean (SO) measured by BGC-Argo float WMO 6901493 (left column) and modelled by SOCA-light-2025 (right column). The variables in each subplot are specified by text in the corresponding subplots. The black stars indicate the depth at which instantaneous PAR value = 15 µmol quanta m^{-2} s^{-1}.

Eastern Mediterranean Sea

The EMS is also a permanent oligotrophic system at temperate latitudes. The float selected (WMO = 6901773) measured all four light variables (PAR, ED380, ED412, and ED490) for nearly four years (Figure 28). Again, the multi-year vertical sections of these variables from this region show very good agreement between the measured and modelled values. The modelled variables exhibited seasonal fluctuations in their magnitude across different years, similar to those observed. The surface incoming solar radiation shows larger seasonal variability than the variability observed in the subtropical oligotrophic regime (NASTG, Figure 26), yet it is well captured by the model. As for the NASTG, the models reproduce light variables with much less noise compared to their corresponding BGC-Argo measurements.

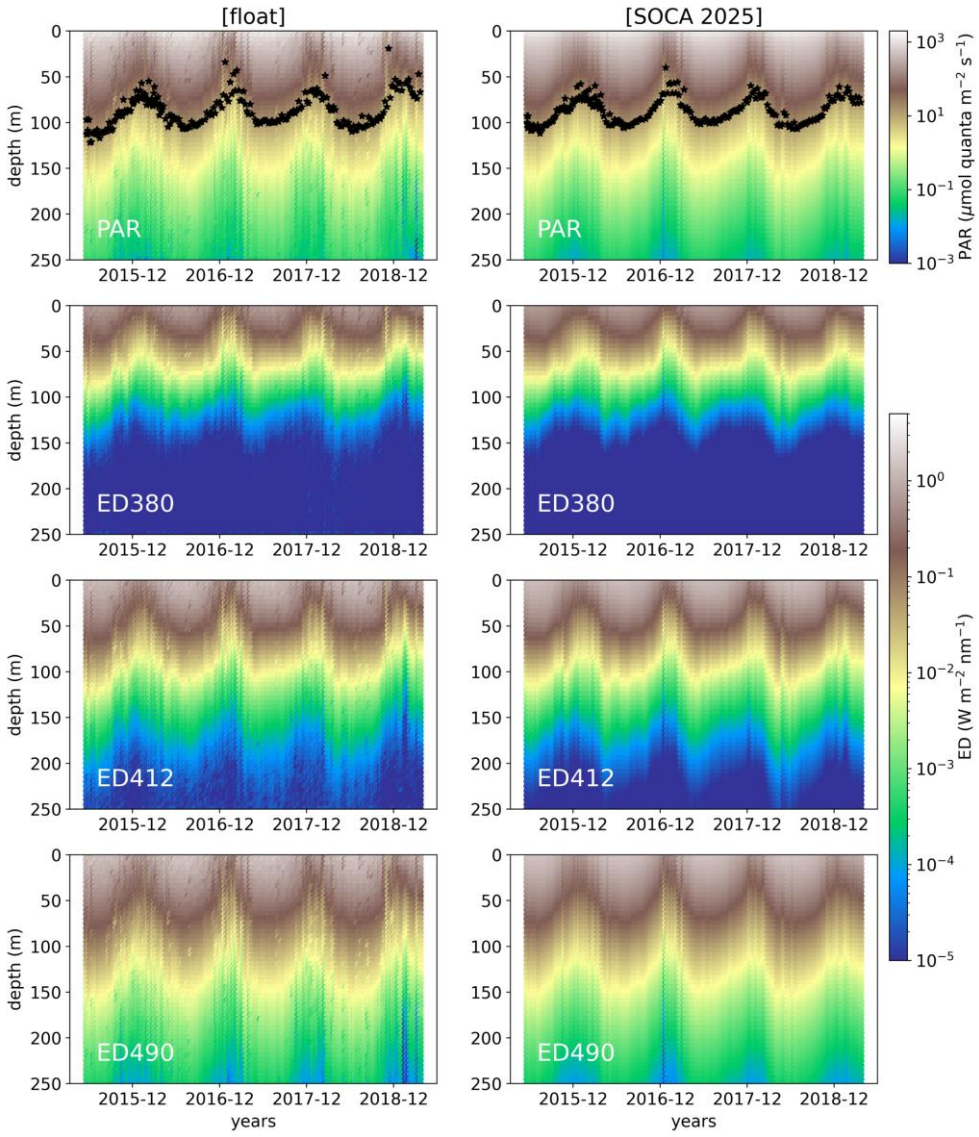


Figure 28: Time series of the vertical distribution of the four light variables in the Eastern Mediterranean Sea (EMS) measured by BGC-Argo float WMO 6901773 (left column) and modelled by SOCA-light-2025 (right column). The variables in each subplot are indicated by text in the corresponding subplots. The black stars indicate the depth at which instantaneous PAR value =15 µmol quanta m^{-2} s^{-1}.

North Atlantic Subpolar Gyre

The data acquired by the float (WMO = 6901523) over its two years of exploration are representative of the diversity of the North Atlantic Subpolar Gyre conditions. In particular, it encountered intense convection periods (>1000 m) as well as intense spring phytoplankton blooms. Due to a lack of ocean colour matchups, the NASPG region experienced similar problems as the SO region in obtaining SOCA-light-2025 variables during the winter. The two-year time series of the vertical distribution of light variables (PAR, ED380, ED412, and ED490) measured by this float and modelled by SOCA-light-2025 are presented and compared in Figure 29. Essentially, the SOCA-light-2025 models reproduce the seasonal trends in float measurements in a smoother way.

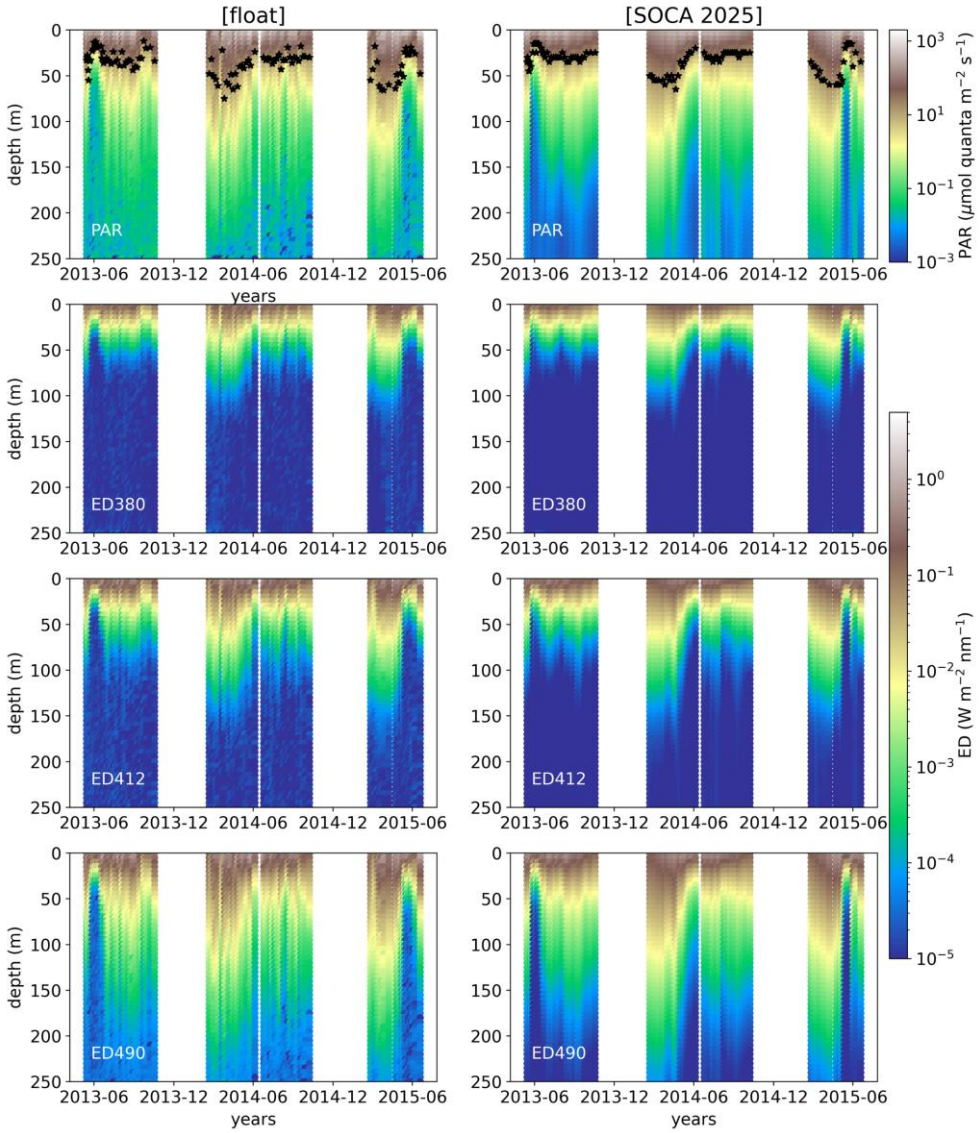


Figure 29: Time series of the vertical distribution of the four light variables in the North Atlantic Subpolar Gyre (NASPG) measured by BGC-Argo float WMO 6901523 (left column) and modelled by SOCA-light-2025 (right column). The variables in each subplot are specified by text in the corresponding subplots. The black stars indicate the depth at which instantaneous PAR value =15 µmol quanta m^{-2} s^{-1}.

IV.3.3 Validation of SOCA-light-2025 models with the independent global SeaBASS database

In addition to validating the SOCA-light-2025 models using 20% of the BGC-Argo dataset and selected BGC-Argo floats excluded from both training and validation phases, we also conducted complementary validations using external datasets. Specifically, we utilized the global SeaBASS light database, which compiles measurements from various research cruises and field campaigns (Werdell et al., 2002, 2003). Unlike the BGC-Argo data, which are collected under all sky conditions, SeaBASS measurements being ship-based and more operator dependent are generally acquired under clear skies.

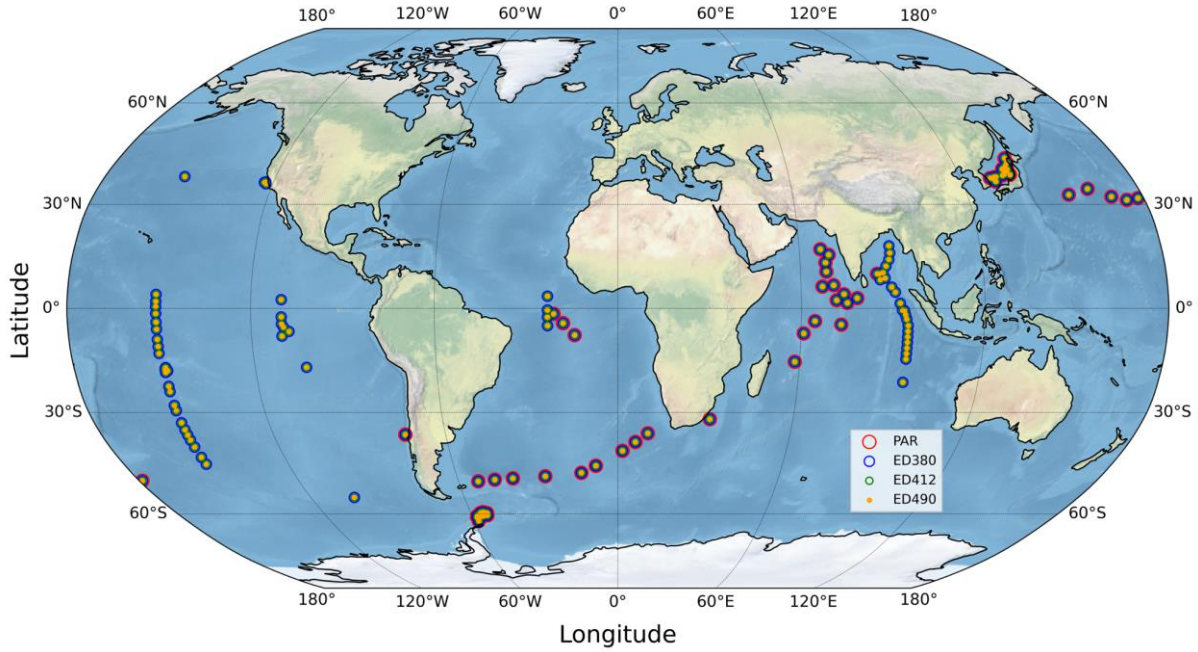


Figure 30: Geographical distribution of independent light-variable profiles (PAR, ED380, ED412, and ED490) available for validation from the SeaBASS database. Red circles represent locations of PAR profiles, blue circles correspond to ED380 profiles, green circles to ED412 profiles, and orange circles to ED490 profiles.

To perform the validation, we matched SeaBASS *in situ* stations with weekly binned data from ARMOR3D (Guinehut et al., 2012; Mulet et al., 2012) and GlobColour. Physical variables (temperature, salinity, and mixed layer depth) were extracted from the nearest 0.25° × 0.25° macro-pixel, while ocean colour data were averaged from a 3 × 3 array of 4 km × 4 km micro-pixels cantered at each station. Due to limitations in satellite data availability mostly caused by signal contamination from clouds or sea ice over 90% of the original SeaBASS stations, including many in shallow coastal regions (depth < 1500 m), were excluded from this validation (Figure 30). The scatterplots of the light variables derived by the SOCA-light-2025 models compared with those measured *in situ* within the SeaBASS database are presented in Figure 31. Figure 31 shows that SOCA-light-2025 retrieval systematically underestimates the SeaBASS measurements for each light variable. This bias is the same over the whole water column as the slopes between the modelled light and corresponding measurements are close to one (Figure 31). The fact that the SeaBASS database is essentially populated by data obtained under clear-sky conditions at a given time could explain this bias. By way of contrast, the weekly matchups of GlobColour products used as input for the SOCA-light-2025 models likely do not correspond to clear-sky conditions over such an extended temporal window. These four light models (PAR, ED380, ED412, and ED490) were validated

independently, and the extracted error metrics are quite satisfactory, even if these statistical estimators are slightly larger compared with the error metrics of both the global 20% validation database and four independent BGC-Argo floats. These larger error estimators could be because of the uncertainty associated with the physical and ocean colour data considered as inputs (as well as the nature of the data in SeaBASS, essentially acquired under clear-sky conditions).

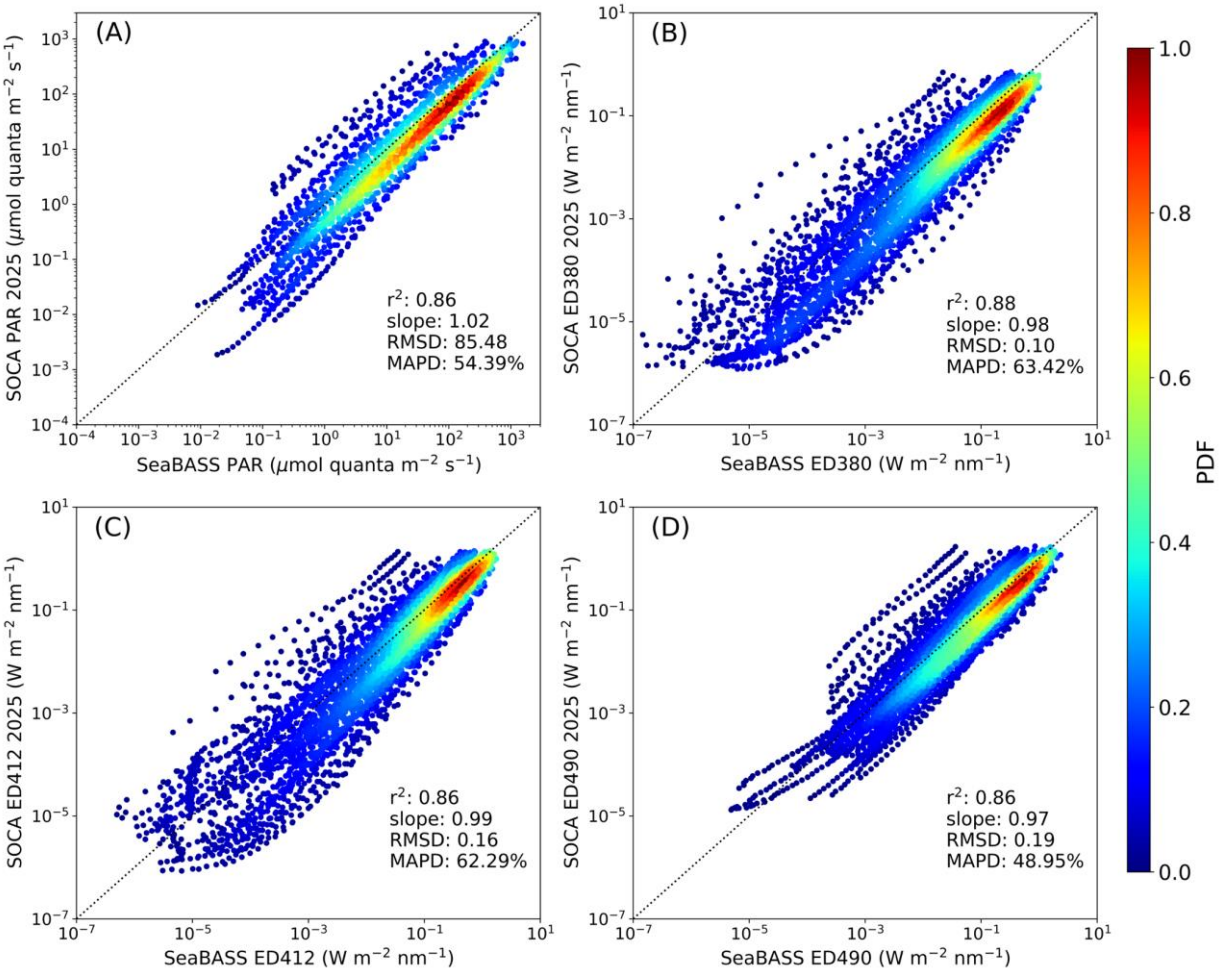


Figure 31: Scatterplots between light variables (PAR, ED380, ED412, and ED490) derived using SOCA-light-2025 models and SeaBASS in situ measurements. The subplots display: PAR (A), ED380 (B), ED412 (C), ED490 (D). The colour code scales the PDF. The identity line is represented by the 1:1 black dotted line.

IV.3.4 Additional validation with iPAR_15

As an alternative to validating SOCA-light-2025 model outputs against external light datasets, another effective approach is to evaluate model-derived products that are critical for specific applications. One such product is the depth of iPAR_15 (**Z_iPAR_15**), defined as the depth at which instantaneous photosynthetically available radiation (iPAR) falls to 15 µmol quanta m^{-2} s^{-1}. This metric is essential for correcting non-photochemical quenching (NPQ), a photo-physiological process that reduces chlorophyll-a fluorescence signals under high light conditions, particularly around midday. The correction method developed by Xing et al., 2018 and refined by Terrats et al., 2020 uses **Z_iPAR_15** as a threshold below which NPQ is negligible. Because **Z_iPAR_15** reflects water clarity, higher values indicating clearer water

and deeper NPQ effects, it serves as a useful proxy for optical water properties. In this study, **Z_iPAR_15** was estimated both from PAR measurements collected by BGC-Argo floats and from PAR simulated by the SOCA-light-2025 model, using the 20% validation subset and four independent floats (Figure 32). The SOCA-light-2025 model showed good performance in retrieving **Z_iPAR_15**. Moreover, the range of **Z_iPAR_15** values observed in the four independent floats (Figure 32B) closely matched that of the broader 20% validation dataset (Figure 32A). This confirms that the four floats represent the full spectrum of trophic conditions captured by the global BGC-Argo dataset.

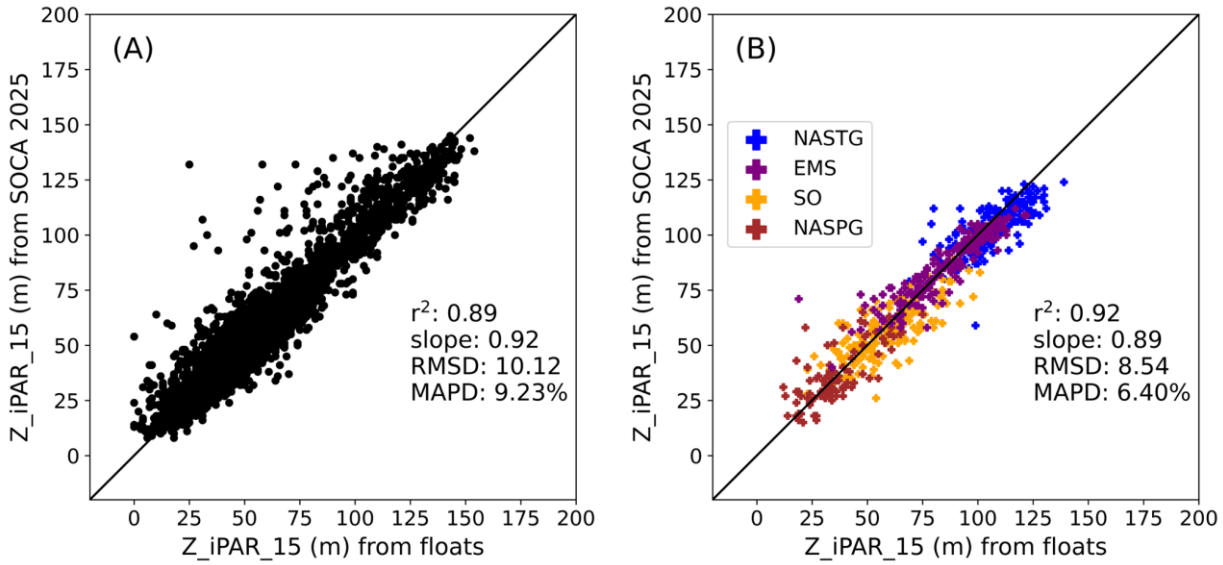


Figure 32: Comparisons of Z_iPAR_15 derived by the SOCA-light-2025 PAR model versus Z_iPAR_15 estimated by BGC-Argo float measurements for the 20% validation database (A) and for the 4 independent floats (B).

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. In includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of *in situ* platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality.

Version 1.0: An issue must be reported concerning the POC variable released for version 1.0. The POC variable was biased because of an error in the processing chain of input data used in the SOCA2020 neural network to estimate the b_{bp} vertical distribution. The water-leaving reflectances have been used as inputs for the training instead of remote sensing reflectance (used to derive the 3D products), biasing the model by a factor of π. The remote sensing reflectance signal (R_{rs}, in sr^{-1}) is multiplied by π to obtain the water-leaving reflectance (Rhow, dimensionless). This led to a non-systematic bias in the POC variable at a global scale, mainly underestimating POC. This issue has been resolved and the unbiased product is released in version 2.0.

Version 2.0: The POC vs. b_{bp} relationship used to derive POC from SOCA2020- b_{bp} was modified for version 2.0. The relationship used in Version 1.0 was the one developed by (Cetinić et al., 2012). The relationship used in Version 2.0 is: $POC = 38687.27 \times {b_{bp}}^{0.95}$

Version 3.0: In version 3.0, 3D Chl product from Uitz et al. (2006) parameterization was replaced by the neural network based SOCA2021 product.

Version 4.0: In version 4.0, the updated SOOCA2024 method for b_{bp} and Chl retrieval was used. In addition to the newly available BGC-Argo data used for training, Chl data has been processed to improve the accuracy. Moreover, the POC vs. b_{bp} relationship has been updated.

Version 5.0: With the release of version 5.0, the dataset now includes enhanced capabilities to represent 3D light profiles. This is achieved through the introduction of variables for PAR, ED380, ED412, and ED490. These light fields are specifically calculated for local noon conditions using the SOCA-light-2025 models (Renosh et al., 2023). The Chl has been updated by using the new dataset available from https://doi.org/10.17882/102324.

Version	Change/Event description
1.0	Issue in the processing of R_{rs} inputs leading to biased b_{bp} and POC estimations
2.0	Replacement of the POC vs. b_{bp} relationship
3.0	Replacement of Uitz et al. (2006) parameterization with the new SOCA2021 neural network for the estimation of Chl
4.0	Major improvement in the Chl database used for the updated SOCA2024 neural network, resulting in significant changes in the estimation of Chl
5.0	The product has been updated to include new light variables (PAR, ED380, ED412, and ED490), derived at local noon. These variables were produced using the SOCA-light-2025 models, which were upgraded in 2025 with an expanded profile dataset and refined inputs (Renosh et al., 2023).

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

As noted above, an issue has been resolved concerning the POC variable distributed in version 1.0 of the product, increasing the quality of the POC product for version 2.0. Moreover, the new POC vs. b_{bp} relationship used to derive POC from SOCA2020- b_{bp} increases significantly the quality of the POC product for the version 2.0.

The quality of the Chl product is considerably improved in the version 3.0 compared with the estimations from the previous Uitz et al. (2006) parameterization method. The quality of the product has been tested using an extensive HPLC database. The statistics of retrieval are significantly better for SOCA2021 while comparing to the former used Uitz et al. (2006) parameterization.

The quality of the Chl product is considerably improved in the version 4.0 compared with the estimations from the previous SOCA2024 method. The product quality has been tested using an extensive HPLC database. The statistics of retrieval are significantly better for SOCA2024 compared with the former SOCA2021 method, especially in the Southern Ocean (Figure 22).

Version 5.0 introduces no quality changes to the existing b_{bp}, and POC variables. The Chl is improved in this version by using the new dataset available from https://doi.org/10.17882/102324 (Version 02-2025) where an issue was fixed in relation to the overestimation of Chl near to the surface in oligotrophic conditions (due to a bug in the MLD computation for some extreme conditions). This version includes new light variables (PAR, ED380, ED412, and ED490), which have been added to the dataset with the same temporal and spatial resolutions.

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